A Multifactor Harmonized Vegetation Index Compositing (MHVC) Approach for Monitoring Vegetation Dynamics in Sandy Land Ecosystems With Landsat-8/9 and Sentinel-2

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***Abstract*—Accurate monitoring of vegetation dynamics in optically complex sandy land ecosystems requires high-spatiotemporal-resolution time series, yet generating such records from individual sensors remains fundamentally constrained by cloud contamination, dust interference, and temporal gaps. However, harmonizing Landsat-8/9 and Sentinel-2 data remains challenging because of cross-sensor spectral inconsistencies, phenological biases, and severe over-masking of valid observations by sand dust and thin clouds, issues that are particularly acute in sandy land ecosystems. Although the harmonized Landsat-Sentinel (HLS) v2.0 product incorporates bidirectional reflectance distribution function (BRDF) normalization and bandpass adjustment, it still exhibits substantial bias in sandy environments (RMSE = 0.14) due to high surface albedo, sparse vegetation, and frequent dust aerosols that amplify anisotropic reflectance and mixed-pixel effects. To address these limitations, we developed the multifactor harmonized vegetation index compositing (MHVC) framework and applied it to the Hulunbuir sandy land from 2017 to 2024. MHVC incorporates three methodological advances: 1) an XGBoost-based spectral harmonization model that removes systematic cross-sensor biases (R^2 = 0.94, *RMSE* = 0.03); 2) a distance-probability joint scoring mechanism that replaces fixed geometric cloud penalties with confidence-modulated weighting, substantially increasing valid pixel retention under dust- and aerosol-prone conditions; and 3) an optimized 40-day compositing window combined with Savitzky–Golay (S-G) filtering for temporally continuous trajectory reconstruction. The framework enables bidirectional gap-filling between sensors with symmetric radiometric fidelity (R^2 = 0.915, *RMSE* = 0.031). Validation against high-quality reference observations demonstrated strong spectral consistency (R^2 = 0.903, *RMSE* = 0.049, *Bias* = 0.001) and enhanced temporal stability, with the median temporal roughness index (TRI) reduced from 0.20 (raw observations) to 0.06 (MHVC), outperforming the multifactor weighted (MFW) baseline (median TRI ≈ 0.08). Independent validation against MODIS MOD13Q1 further confirms phenological consistency between MHVC and the coarser-resolution benchmark (r = 0.888), with residual discrepancies attributable to the inherent scale difference between 30- and 250-m products rather than to framework errors. Cross-regional application to the Horqin, Hunshandak, and Mu Us sandy lands yield consistently high spectral fidelity (R^2 = 0.882–0.935) and temporal stability (median TRI reduction confirmed in three of four regions), confirming the framework's transferability without region-specific recalibration. Overall, MHVC provides a robust, phenologically faithful solution for generating multiyear vegetation time series in sandy land ecosystems, offering direct applicability to desertification monitoring and ecological management in optically challenging drylands across northern China and a broader potential for vegetation dynamics studies across global sandy lands, shrublands, and other sparsely vegetated regions where optical complexity limits conventional time-series approaches.**

***Index Terms*—Data harmonization, Landsat-8/9, Multifactor Harmonized Vegetation index Compositing (MHVC), sandy land, Sentinel-2.**

I. Introduction

SANDY ecosystems encompass approximately 20% of the Earth's land surface and underpin global environmental sustainability through water conservation, biodiversity support, and climate regulation [1], [2]. Characterized by sparse vegetation and strong spatial heterogeneity [3], [4], these regions face disproportionate exposure to disturbances from dust storms, extreme climate events, and human activities [5], [6], [7]. Timely and continuous monitoring of vegetation dynamics in sandy lands is therefore essential for effective ecological management [8], [9], [10]. Vegetation indices (VIs) are widely used for this purpose due to their simplicity, robustness, and strong association with vegetation conditions [11], [12]. However, in sandy environments with high soil background reflectance, sparse canopy cover, and pronounced spatial variability [13], single-sensor observations are often compromised by soil contamination and insufficient temporal density to resolve rapid phenological changes, making it difficult to achieve both sufficient spatial detail and sufficient temporal continuity [14].

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Although Landsat and Sentinel-2 are among the most widely used satellite datasets, both present well-documented constraints in sandy land monitoring. Landsat-8 provides a 30-m spatial resolution sufficient to capture fine-scale sandy land features, but its 16-day revisit interval [15], [16] and occasional data gaps [17], [18] often lead to substantial temporal discontinuities during the growing season [16], [19]. Sentinel-2 offers improved temporal sampling with a five-day revisit cycle [15], [20]; however, under adverse conditions [21], [22], both sensors can be simultaneously affected, producing gaps or systematic shifts in the VI time series [16], [21]. Integrating these two datasets into a virtual constellation [20], [23] offers the potential for high-spatiotemporal-resolution imagery [17], [24]. Although recent advances such as harmonized Landsat-Sentinel (HLS) v2.0 have improved cross-sensor harmonization through bidirectional reflectance distribution function (BRDF) normalization and bandpass adjustment, these linear or semi-empirical approaches still struggle in sandy lands where high albedo, sparse canopy, and frequent dust events create highly nonlinear spectral and angular discrepancies. Specifically, the high surface albedo of sandy substrates amplifies anisotropic reflectance across the swath in ways that a globally invariant linear bandpass adjustment coefficient cannot adequately capture [17]; sparse vegetation canopies produce spectrally mixed pixels whose composition varies with viewing geometry in a manner that departs substantially from the global average BRDF model parameters derived from MODIS MCD43A1 [17]; and frequent dust aerosol loading introduces spectrally correlated offsets across all bands that exceed the correction capacity of the LaSRC atmospheric correction, which has been shown to produce its largest residual errors under high aerosol optical thickness conditions characteristic of arid and semi-arid regions [17]. Machine learning approaches, by contrast, can learn these complex multivariate interactions directly from co-located observations, providing a principled path to overcoming these limitations [25], [26]. These considerations collectively motivate a departure from fixed-coefficient linear correction toward an adaptive, nonlinear harmonization strategy that can learn environment-specific spectral relationships directly from co-located observations [27].

Beyond cross-sensor correction, constructing stable and temporally consistent VI time series from cloud-prone, data-sparse observations remains a major challenge [27]. Early compositing approaches, such as maximum value composite (MVC) and maximum NDVI composite (MNC), mitigated the effect of cloud contamination by selecting the highest NDVI value within a given time window [28], [29]. However, these methods are prone to errors in nonvegetated areas, densely vegetated zones, or highly reflective sandy surfaces [18], [30]. To address these limitations, subsequent studies introduced multicriteria scoring methods. Griffiths et al. [31] proposed a pixel-based compositing approach that integrates acquisition timing, day of year (DOY), and distance to clouds into a unified scoring framework. White et al. [32] extended this concept with the best-available-pixel (BAP) method, incorporating sensor priority and atmospheric conditions to improve pixel selection across multiple sensors. Griffiths et al. [33] further developed a weighted parametric scoring method based on a weighted linear combination of the individual scores. Frantz et al. [34] subsequently introduced the phenology adaptive compositing (PAC) method, which dynamically adjusts the target date based on local phenological conditions, enabling more temporally coherent pixel selection. A comprehensive evaluation of ten image compositing algorithms demonstrated that multirule methods, including pixel-based compositing, BAP, and weighted parametric scoring, produced reliable annual composites in spectral and land cover classification assessments [35], with the BAP algorithm outperforming others in spatial assessment across all compositing periods [18]. More recent approaches allow criterion weights to be adaptively adjusted based on local conditions, enhancing compositing flexibility [30], [36]. However, because the BAP method selects the single highest-scoring pixel, observations across different locations may originate from different seasons [32], potentially introducing substantial radiometric inconsistencies driven by phenological variation [35], [36].

Despite these advances, best-pixel selection methods such as MVC and BAP share a fundamental limitation: selected pixels often originate from different phenological stages, which introduces inconsistencies into the composite images [32], [35]. This issue is particularly pronounced during the short and rapidly changing growing season in sandy lands [4], [37]. To address this shortcoming, Meng et al. [36] proposed the multifactor weighted (MFW) compositing method. In this approach, Sentinel-2 imagery is first converted to OLI-like datasets to ensure cross-sensor spectral consistency [38]. Cloud and cloud shadow pixels are then identified, and topographic effects on image radiometry are corrected [39], [40]. Three scoring indicators are subsequently computed: acquisition DOY, distance of each pixel to the nearest cloud or cloud shadow pixel, and haze impact level [36]. The final composite is derived through a weighted averaging approach in which the weight assigned to each observation is defined as the sum of its three scores [36]. Unlike best-pixel selection methods such as BAP, this method improves spatial continuity and radiometric consistency by integrating multiple observations rather than relying on a single pixel [31], [34]. Evaluations across four study sites spanning diverse climate zones in China demonstrate that MFW composites achieve higher spectral agreement with reference images (R = 0.78–0.95) compared to BAP composites (R = 0.65–0.93) [36]. By leveraging multiple observations, the MFW approach enhances temporal smoothness and reduces the influence of outliers, representing a significant advancement in time-series compositing [35].

However, the MFW method is subject to two notable limitations. First, its performance is highly sensitive to the width of the DOY weighting window. In areas with few available observations, a wider window is required to ensure sufficient data availability; however, this often introduces observations from different phenological stages, thereby reducing temporal consistency, particularly during the short growing season in sandy regions [4], [41]. Second, the conventional cloud-distance scoring component relies on binary cloud masks

and simple distance measurements, which systematically tends to misidentify thin clouds and dust aerosols as thick cloud cover [40], [42]. This issue is especially pronounced in sandy lands, where high surface reflectance can lead to false cloud detections [43], [44]. As a consequence, many valid, clear pixels near cloud edges receive low scores and are excluded, further reducing data availability [36]. Together, these limitations constrain the ability of existing methods to generate stable VI time series during extended periods of missing data [41], [45].

To address these limitations, this study proposes an improved cloud-distance scoring approach that replaces binary cloud masks with continuous cloud probability values derived from an independent cloud probability layer [44], [46]. Rather than applying a fixed penalty-based solely on distance to the nearest detected cloud, the revised method adjusts the effective distance according to the cloud probability of the nearest flagged pixel [39]. This approach attenuates penalties for pixels near uncertain or low-confidence cloud detections, allowing more likely clear observations to receive higher scores [36]. By recovering more usable observations during data-sparse periods, this modification improves the stability and consistency of the compositing framework [18], [35].

Complementing compositing strategies, temporal reconstruction methods can further enhance time-series continuity [47], [48]. Common approaches include interpolation based on logistic growth models and harmonic analysis, as well as smoothing filters such as the Savitzky–Golay (S-G) filter [49], [50]. These methods generally work well when the input series is relatively complete and clean [51], [52]. However, when preliminary composites contain frequent gaps or high noise levels, as is often the case with single-sensor Landsat-8 or Sentinel-2 data in sandy regions [37], [41], their performance declines significantly [53], [54]. This limitation has promoted the development of combined strategies that first fill gaps through interpolation and then apply smoothing filters [52], [55]. Such approaches are better suited to the irregular, multisource time-series typical of medium-resolution satellite observations [19], [45].

Against this background, this study presents the multifactor harmonized VI (Harmonized-VI) compositing (MHVC) framework, designed to generate VI time series in sandy land environments with both high spatial detail and strong temporal continuity. The framework integrates three key components, cross-sensor spectral correction, multifactor image compositing, and temporal reconstruction, into a unified workflow to enhance the quality and accuracy of VI data in sandy regions. The specific objectives are listed.

1) To build a nonlinear machine learning-based mapping model between Landsat-8/9 and Sentinel-2 VIs to correct systematic differences caused by sensor-specific spectral response functions (SRFs) prior to compositing.
2) To design an improved cloud and shadow scoring method that incorporates both cloud proximity and cloud probability, thereby reducing the over-penalization of valid observations affected by thin clouds and dust aerosols in sandy environments.

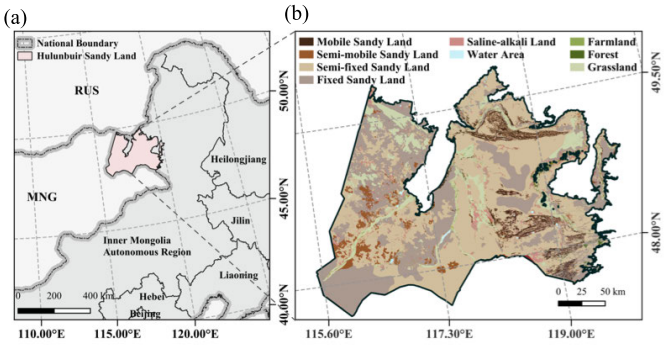


Fig. 1. (a) Location of the study area. (b) Distribution of sandy land types.

3) To determine the optimal DOY weighting window for sandy land conditions through systematic evaluations, and to apply a Gaussian-weighted S-G filter for temporal reconstruction to reduce residual noise and fill data gaps while preserving key phenological features such as rapid vegetation green-up.

The MHVC framework proposed in this study addresses a critical gap in vegetation monitoring for optically complex, data-sparse sandy environments. By systematically resolving cross-sensor inconsistency, cloud and aerosol contamination, and temporal discontinuity within a unified workflow, this framework enables the generation of VI time series that are both spatially detailed and phenologically consistent, qualities that remain challenging to achieve with existing methods in sandy land contexts. The resulting outputs can directly support large-scale ecological assessments, desertification monitoring, and the evaluation of restoration effectiveness in these regions. Furthermore, the methodological advances developed in this study, particularly the probability-aware cloud scoring and the adaptive DOY weighting strategy, offer transferable insights for medium-resolution time-series construction in other dryland and semi-arid environments where dust aerosols, limited observations, and rapid phenological dynamics present similar challenges. The framework's generalizability is further demonstrated through application to the Horqin, Hunshandak, and Mu Us sandy lands, confirming robust performance across the full hydroclimatic gradient of temperate sandy ecosystems in northern China.

II. Study Area and Data

A. Study Area

The Hulunbuir sandy land is located in the northeastern part of the Inner Mongolia Autonomous Region, China, situated in the agro-pastoral ecotone of the northern frontier [Fig. 1(a)]. Covering approximately 47 600 km^2, it is characterized by a sandy, sparse-forest grassland landscape [37]. Unlike the hyper-arid sandy systems that dominate the global dryland remote sensing literature, the Hulunbuir sandy land is a high-latitude, semi-arid to semi-humid ecosystem under active vegetation degradation and recovery dynamics, a category that remains comparatively underrepresented in existing studies. Several characteristics make it a particularly demanding testbed for multisource remote sensing. First, its location at approximately 47°–50° N produces a short,

phenologically compressed growing season (April–October) with rapid and large-amplitude NDVI transitions, making time-series reconstruction from cloud-interrupted observations especially challenging. Second, the region spans a continuous spectral gradient from near-bare mobile dunes to recovering semi-fixed grassland and sparse forest, with NDVI values ranging from near zero to above 0.6 within distances of a few kilometers; this broad dynamic range imposes stringent demands on sensor harmonization and compositing algorithms simultaneously. Third, cloud contamination is concentrated in the peak growing season (June–August), precisely when the vegetation signal is strongest, making temporal gap-filling both more severe and more ecologically consequential than at lower latitude dryland sites.

As shown in the land cover classification [Fig. 1(b)], the region is a complex mosaic of fixed and semi-fixed sandy lands interspersed with patches of forest, grassland, and agricultural fields. This high surface heterogeneity creates a significant "mixed pixel" challenge for remote sensing: the weak spectral signals of sparse vegetation are often masked by the bright sandy background [56], [57], often leading to the underestimation of NDVI in moderate-to-low coverage areas [58], [59]. The 30-m spatial resolution of Landsat-8/9 is sufficient to distinguish the dominant landscape units of the Hulunbuir sandy land, including fixed dune patches, semi-fixed grassland, and agricultural fields, whose characteristic spatial scales range from approximately 50 m to several kilometers; however, subpixel mixing remains a challenge for isolated shrubs and narrow vegetation-dune transition zones, which motivates the multiscale spatial feature design described in Section III-A. Therefore, the Hulunbuir sandy land serves not only as an ideal testbed for developing and validating the proposed MHVC framework, but also as a representative proxy for the broader class of high-latitude, seasonally cloud-affected sandy ecosystems that remain relatively underrepresented in existing medium-resolution time-series studies.

B. Remote Sensing Data

This study collected all available Landsat-8/9 OLI and Sentinel-2 MSI images covering the study area from 2017 to 2024. To focus on vegetation dynamics and minimize snow interference, data acquisition was limited to the growing season (April 1–October 31) of each year [60]. The Landsat-8/9 data were obtained from the USGS EarthExplorer, utilizing the Collection 2 Level-2 surface reflectance product (C2 L2 SR). These data offer a 30-m spatial resolution and a 16-day revisit cycle per satellite, resulting in an effective eight-day revisit interval when Landsat-8 and 9 are used in combination [61], [62], [63]. Atmospheric correction was performed using the LaSRC algorithm, providing spectral coverage from the visible to the shortwave infrared [62], [64]. Sentinel-2 data were acquired from the Copernicus Data Space Ecosystem, utilizing the Level-2A (L2A) surface reflectance product processed with the Sen2Cor atmospheric correction algorithm [65]. Sentinel-2 provides 13 spectral bands at multispectral spatial resolutions of 10, 20, and 60 m, with a five-day revisit interval sustained by its twin-satellite constellation [66].

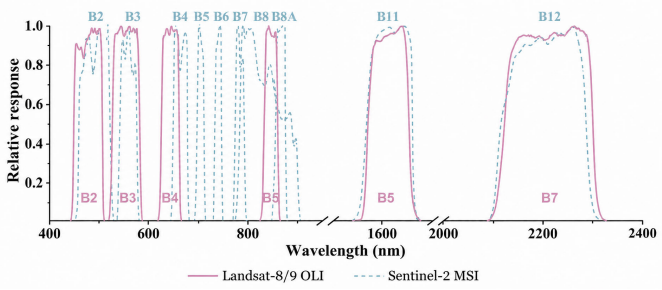


Fig. 2. Comparison of SRFs between Landsat-8/9 OLI (solid pink lines) and Sentinel-2 MSI (dashed blue lines) for spectrally corresponding band pairs. The band labels above each curve denote the Sentinel-2 MSI band designations (e.g., B2, B3, B4, B5/B6/B7/B8/B8A, B11, and B12), while the labels within the plot area denote the corresponding Landsat-8/9 OLI bands (B2–B7). The systematic offsets in peak wavelength and bandwidth between the two sensors motivate the cross-sensor harmonization described in Section III-A.

Although both Landsat-8/9 OLI and Sentinel-2 MSI provide atmospherically corrected surface reflectance data, systematic differences in their spectral and geometric characteristics must be addressed prior to data compositing [17], [38], [61], [67]. The two sensors have distinct SRFs for nominally corresponding bands, which can introduce systematic radiometric biases for the same surface type (as shown in Fig. 2) [17], [68], [69], [70]. In addition, differences in geometric correction and atmospheric processing pipelines between the two sensors can cause spatial misregistration, compromising the temporal and spatial consistency of the composite time series [17], [45], [61], [71]. Consequently, cross-sensor radiometric harmonization and geometric co-registration are essential prerequisites for reliable integration of the two datasets [17], [67], [68], [69].

III. Methodology

The proposed MHVC framework consists of three main stages: 1) machine learning-based cross-sensor harmonization; 2) multifeature weighted compositing (MFW) for time-series reconstruction; and 3) accuracy assessment. The overall technical roadmap is illustrated in Fig. 3.

As shown in Fig. 3, the framework proceeds through four sequential processing stages. In the first stage, Landsat-8/9 OLI and Sentinel-2 MSI L2A surface reflectance images are resampled to a common 30-m spatial grid and co-registered to ensure pixel-level consistency prior to spectral processing. In the second stage, an XGBoost-based harmonization model corrects cross-sensor spectral discrepancies using a multicategory feature set that encodes spectral, geometric/angular, spatial context, and temporal information, thereby accounting for BRDF-induced reflectance offsets and mixed-pixel effects at sand–vegetation boundaries. In the third stage, harmonized observations from both sensors are integrated through the improved compositing scheme: each candidate pixel receives a quality score based on its temporal proximity to the target date [S_{DOY}, (1)], cloud contamination level [$S_{\text{CloudDist}}$, (3)], and haze impact [S_{HOT}, (5)], and the final composite value is derived as their weighted average [see (7)]. In the fourth stage, a Gaussian-weighted S-G filter is applied to the resulting eight-day composite time series to suppress residual noise, yielding

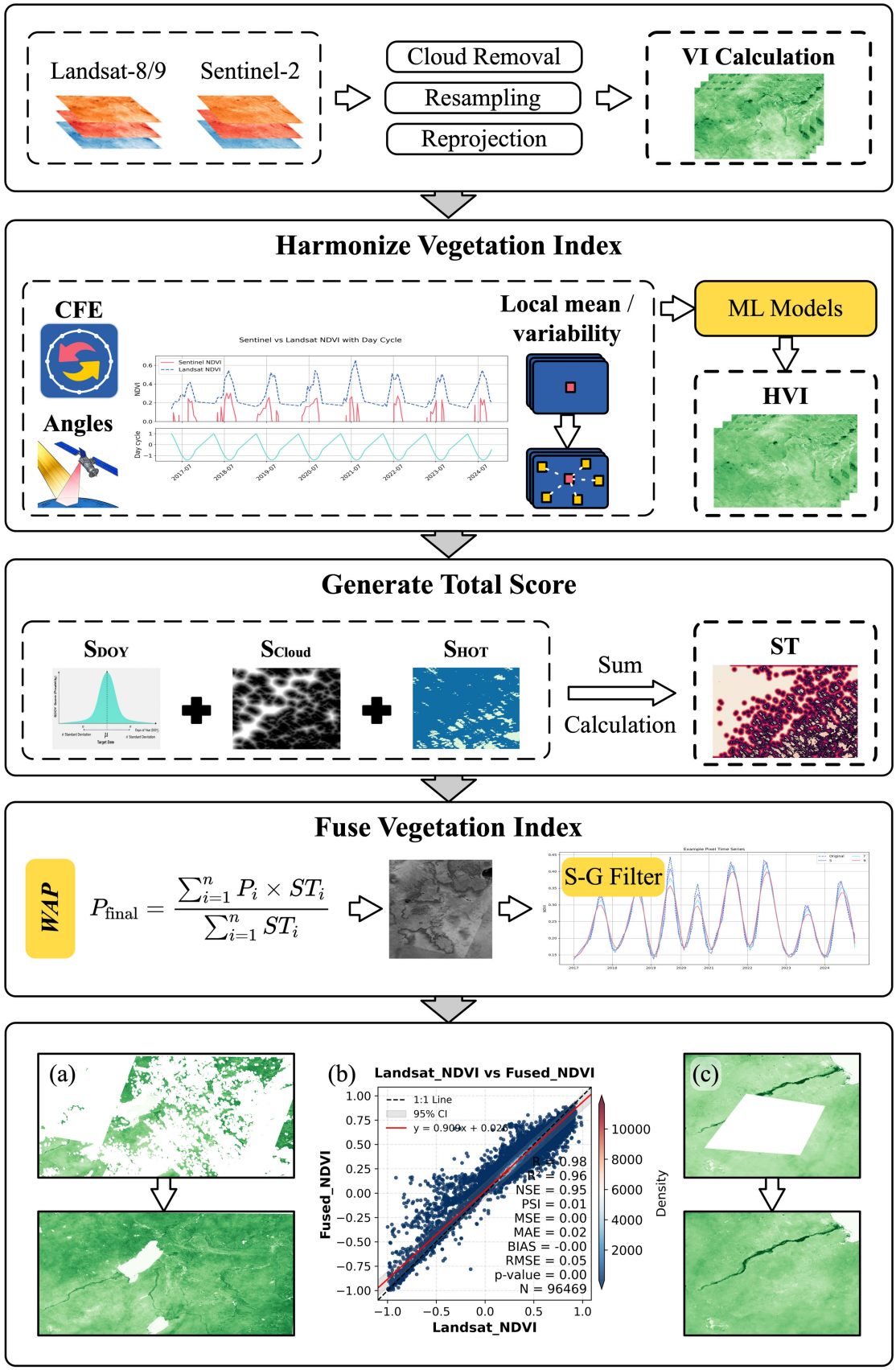


Fig. 3. Technical roadmap of the proposed MHVC framework. (a) Gap-filling effect of MHVC in scenes with missing data across the study area using Landsat-8/9. (b) Scatter plot showing the correlation between MHVC-reconstructed data and original landsat-8/9 observations. (c) Gap-filling performance of MHVC on artificially created (known) data gaps.

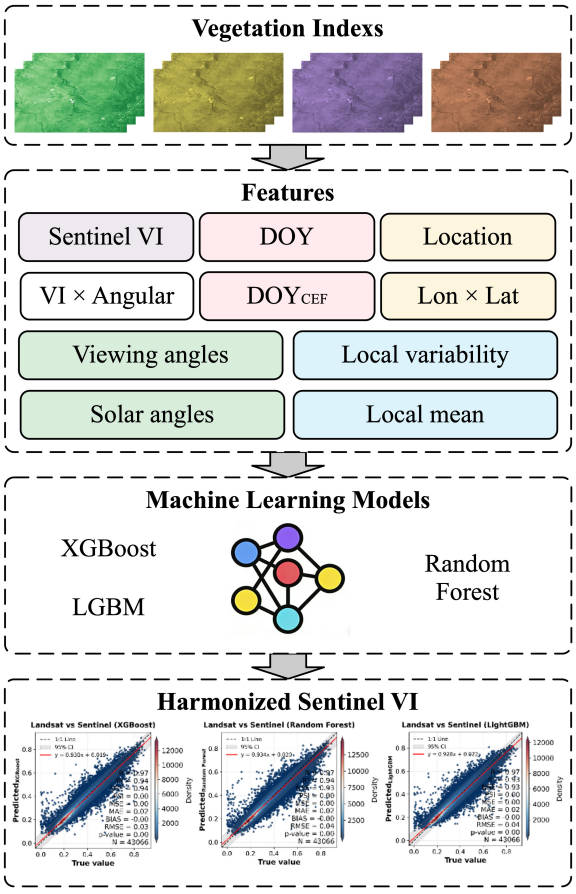


Fig. 4. Model–feature system schematic.

spatiotemporally continuous 30-m VI products for subsequent accuracy assessment.

A. Harmonized VI

To generate high-quality Harmonized-VI data, this study developed a machine learning-based multifeature compositing model to establish a robust mapping between Sentinel-2 and Landsat-8/9 VIs. The model incorporates a comprehensive feature framework, including basic observational parameters, spatial structural representations, temporal dynamic patterns, and composite interaction features (Fig. 4), supplying both a physical rationale and a statistical foundation for cross-sensor correction.

Differences in solar elevation, solar azimuth, and sensor viewing angles during Landsat and Sentinel acquisitions can substantially influence surface reflectance [17], [72], [73]. By incorporating these angular parameters into the model, we effectively account for biases arising from observation geometry [74], a factor that is particularly consequential in the sand–vegetation ecotone where anisotropic reflectance is amplified by bright soil backgrounds.

Spatial features are constructed using a multilevel strategy: geographic coordinates are standardized to create relative position features, mitigating numerical scaling issues caused by regional differences in absolute latitude and longitude values, and enhancing data comparability [45], [59], [75], [76], [77]. Latitude–longitude interaction terms are included to capture nonlinear spatial dependencies. In addition, multiscale spatial statistical features are derived by using sliding windows of 0.1°, 0.2°, and 0.3° to calculate local statistics (mean, standard deviation, maximum, minimum) of the Sentinel VI, effectively simulating spatial autocorrelation patterns at different scales. Together, these multiscale neighborhood descriptors enable the model to disentangle fine-scale patch variability from broader landscape gradients, a distinction that is critical in the highly heterogeneous ecotone environment of the study area.

Temporal features are encoded by transforming the DOY into a continuous cyclic vector using sine and cosine transformations. This approach resolves the year-boundary discontinuity inherent in the conventional DOY representation and enables the model to capture seasonal changes in vegetation phenology more accurately [47], [48], [51].

Composite features emphasize interaction effects between variables. Interaction terms between the VI and observation angles are constructed to represent changes in reflectance under different viewing geometries, thereby correcting atmospheric or observation distortions caused by angle differences [72], [74], [78], [79]. Simultaneously, spatial grouping features are generated by discretizing continuous geographic locations into representative spatial clusters via latitude–longitude quantile partitioning. This approach allows the model to identify region-specific vegetation patterns and supports grouped cross-validation to prevent spatial data leakage, thereby enhancing the model's generalization performance across heterogeneous geographic regions [75], [77].

To interpret the trained XGBoost model and quantify the relative contribution of different features, we calculated the feature importance using the gain metric, which measures the average improvement in model accuracy attributed to each feature across all tree splits where it is employed. The features were then grouped into four categories, spectral, geometric/angular, spatial context, and temporal, to evaluate their respective contributions to cross-sensor harmonization performance [77].

B. Improved Pixel Quality Scoring and Weighted Compositing

After generating the Harmonized-VI, this study applied a modified MFW approach for time-series reconstruction [36]. This method employs a multidimensional pixel-quality scoring system to assign physically motivated weights to each observation, producing a high-quality composite outcome. For the sandy study area characterized by sparse observations, complex cloud contamination, and rapid phenological changes, two key improvements were introduced to the traditional MFW method. First, the DOY scoring window was optimized to minimize the inclusion of observations from different phenological phases. Second, the cloud and cloud shadow score was enhanced by incorporating continuous probability weights from an independent cloud probability layer, thereby attenuating penalties for thin clouds and dust aerosols and increasing the weight of genuinely clear pixels. These modifications effectively increase both the quantity and quality of candidate pixels under sparse observation conditions, improving the temporal consistency and stability of the weighted composites.

1) Score for the DOY: To quantify the temporal consistency between the observation date and the target date, we construct a Gaussian-based date-weight scoring function [36]

$$S_{DOY} = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2\right) \tag{1}$$

where x_i is the date (DOY) of the candidate observation, μ is the target date, and σ is a temporal scale parameter controlling the decay rate of the time weight. This scoring function ensures that observations closer in time to the target date receive higher weights, while those with larger time differences have their weights reduced rapidly [32], [36], [47]. We employ a ±20-day time window (40-days total) to ensure a sufficient number of samples while effectively suppressing inconsistencies due to vegetation phenological changes, thereby improving the temporal continuity and stability of the composite results [48], [51], [80].

2) Score for the Cloud and Cloud Shadow: To quantify the impact of clouds and cloud shadows on pixel quality, we construct a distance–probability joint scoring function [36], [46]

$$D_{\text{effective}} = D_i \cdot (1 - P_{\text{closest}} + \epsilon) \tag{2}$$

$$S_{\text{CloudDist}} = \frac{1}{1 + e^{\left(-0.2 \cdot \min\{D_{\text{effective}}, D_{\text{req}}\} - \frac{D_{\text{req}} - D_{\min}}{2}\right)}} \tag{3}$$

where D_i is the Euclidean distance from the target pixel to the nearest cloud or shadow pixel, $P_{closest}$ is the probability or confidence of that nearest cloud/shadow pixel, ϵ is a buffer constant to avoid over-penalizing when a high-probability cloud results in an effective distance of zero, D_{req} is the distance threshold (set to 50 pixels), and D_{min} is the minimum distance (set to 0 pixels). This method incorporates a probability-weighted effective distance on top of the traditional binary distance, substantially reducing penalties for pixels affected by low-probability thin clouds or other uncertain interference (such as dust) [39], [42], [44], [46]. Consequently, those pixels receive higher scores, more accurately reflecting their usability in the compositing process.

To evaluate the performance of the cloud scoring mechanism, the valid pixel proportion was defined as the percentage of pixels within a scene receiving a quality score above the usability threshold. Comparisons between MFW and MHVC were conducted across three representative scenario categories: Scenario A, pixels affected by thin cloud or dust aerosol with low but nonzero cloud probability (<0.3); Scenario B, pixels located within 10 pixels of a high-confidence cloud boundary where mixed-pixel contamination is likely; and Scenario C, cloud-free sandy pixels erroneously flagged as cloud or shadow by the detection algorithm, identified via cross-referencing with manually verified clear-sky reference images.

3) Score for the HOT: To identify and suppress pixels affected by aerosols, thin clouds, and haze, we use the Haze-optimized transformation (HOT), based on the blue and red bands [36], [43]

$$HOT = \rho_{blue} - 0.5\rho_{red} - 0.08 \tag{4}$$

$$S_{HOT} = \frac{1}{1 + e^{\left(\frac{10}{0.02}(HOT + 0.075)\right)}} \tag{5}$$

where ρ_{blue} and ρ_{red} are the reflectances in the blue and red bands, respectively. The HOT index amplifies the sensitivity of the blue band to aerosols and thin clouds, effectively distinguishing haze-contaminated pixels from clear ones. Its corresponding sigmoid scoring function maps HOT values continuously into weights in [0, 1], significantly lowering the scores of pixels with high haze contamination while maintaining high weights for clear pixels.

4) Image Compositing Based on the Total Score: Unlike traditional BAP methods that select only the single highest-scoring observation, the MFW method fully utilizes all valid observations through weighted averaging [32], [36]

$$ST = S_{DOY} + S_{CloudDist} + S_{HOT} \tag{6}$$

$$P_{final} = \frac{\sum_{i=1}^{n} P_i \times ST_i}{\sum_{i=1}^{n} ST_i} \tag{7}$$

where P_{final} is the composite VI value, and P_i and ST_i are the original value and total score of the ith observation, respectively, with n being the number of available observations. This weighted compositing strategy not only avoids the "patchy" effect that can arise from the BAP method but also enhances the temporal continuity and spatial consistency of the composite results [18], [36].

5) Time-Series Smoothing and Noise Reduction: After generating the improved 30-m MFW-NDVI composite time series with significantly enhanced spatiotemporal completeness, we applied an S-G filter to smooth the VI time series and suppress noise caused by residual cloud detection errors and other factors [52].

The S-G filter, a widely used digital algorithm for relatively continuous VI time series (e.g., MODIS), has been extensively validated [53]. Its principle is to fit local polynomials to the time series, effectively reducing noise while preserving the original phenological signal [55]. Since cloud contamination typically induces a negative bias in NDVI values, we implemented a weighted S-G filter following Chen et al. [49], designed to fit the upper envelope of the NDVI time series [81]. This approach corrects the low-frequency negative bias caused by clouds, producing a high-quality, high-fidelity smoothed NDVI time series as a reliable basis for extracting phenological parameters.

C. Accuracy Assessment and Evaluation Metrics

To rigorously evaluate the effectiveness of the MHVC framework, we implemented a comprehensive validation scheme focusing on two key dimensions: spectral fidelity and temporal stability. These two dimensions are complementary: spectral fidelity quantifies how accurately reconstructed NDVI values match contemporaneous reference observations, while temporal stability captures the degree to which residual noise

and compositing artifacts are suppressed in the resulting time series.

For spectral accuracy, the reconstructed NDVI values derived from the MHVC framework were compared pixel-by-pixel against high-quality Landsat reference observations. A set of widely recognized statistical metrics was employed, including the coefficient of determination (R^2), Pearson correlation coefficient (r), root-mean-square error (RMSE), mean absolute error (MAE), and bias [54], [82]. These metrics are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{8}$$

$$r = \frac{\sum_{i=1}^{n} \left(\hat{y}_i - \bar{\hat{y}}\right)(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} \left(\hat{y}_i - \bar{\hat{y}}\right)^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}} \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2} \tag{10}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_i - y_i| \tag{11}$$

$$Bias = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) \tag{12}$$

where $\hat{y}_i$ and y_i represent the reconstructed and reference NDVI values for the ith pixel, respectively, $\bar{\hat{y}}$ and $\bar{y}$ are their corresponding mean values, and n is the total number of validation pixels.

For temporal consistency, we employed the temporal roughness index (TRI) to quantify the smoothness of the reconstructed NDVI time series [50], [83], [84]. TRI is calculated as the mean absolute difference between consecutive time steps [see (13)]. A lower TRI indicates a smoother and more temporally stable time series, reflecting the effective suppression of residual noise (e.g., from thin clouds or dust aerosols) and the preservation of natural phenological trajectories. TRI values were computed for raw observations, traditional MFW composites, and the proposed MHVC reconstructions to provide a three-way benchmark for assessing the temporal stabilization contribution of each processing stage

$$TRI = \frac{1}{m-1} \sum_{t=1}^{m-1} |NDVI_{t+1} - NDVI_t| \tag{13}$$

where $NDVI_t$ is the VI value at time step t, and m is the total number of composite periods.

IV. Results

A. Visual Assessment

1) Reconstruction Performance Across Phenological Stages: Section IV presents quantitative and visual evidence that the MHVC framework outperforms BAP and MFW across four complementary evaluation scenarios. To evaluate the spatial fidelity of the proposed framework, Fig. 5 presents a visual comparison of local subareas during the peak growing season (June 4, 2024). Overall, all evaluated methods successfully capture the broad-scale spatial patterns of vegetation in the sandy regions, producing composite images with high consistency in hue and brightness relative to the RGB reference [Fig. 5(a)–(c)].

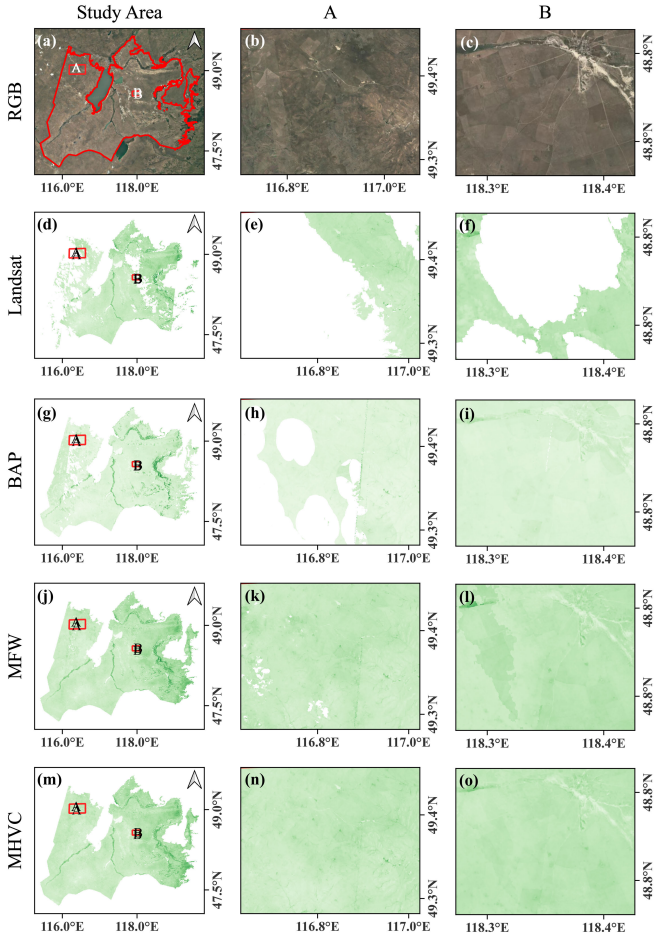


Fig. 5. Visual comparison of NDVI composites using different methods in the Hulunbuir sandy land (Landsat reference acquired on June 4, 2024). (Rows from top to bottom) (a)–(c) RGB reference, (d)–(f) Landsat composite, (g)–(i) BAP, (j)–(l) MFW, and (m)–(o) MHVC.

However, subtle performance differences become apparent in highly heterogeneous regions, such as the transition zones between vegetation and bare sand [Fig. 5(b)–(c) and (n)–(o)]. Due to its pixel-level selection mechanism, the BAP method introduces spectral discontinuities, resulting in localized salt-and-pepper or patchy artifacts [Fig. 5(h)–(i)]. While the MFW method enhances spatial continuity through its weighting approach, it tends to over-smooth object edges and texture details, leading to degraded sharpness in transition zones [Fig. 5(k)–(i)].

In contrast, the proposed MHVC method demonstrates superior preservation of local structures [Fig. 5(n)–(o)]. By integrating temporal constraints with optimized weights, MHVC effectively suppresses the mosaic effects observed in BAP [Fig. 5(h)–(i) versus (n)–(o)] and surpasses MFW in retaining the internal textures of surface features, such as dune ridges and vegetation patch edges [Fig. 5(k)–(l) versus (n)–(o)]. Across the spring, summer, and autumn phenological periods (Fig. 5), MHVC consistently maintains high visual coherence. The resulting images most

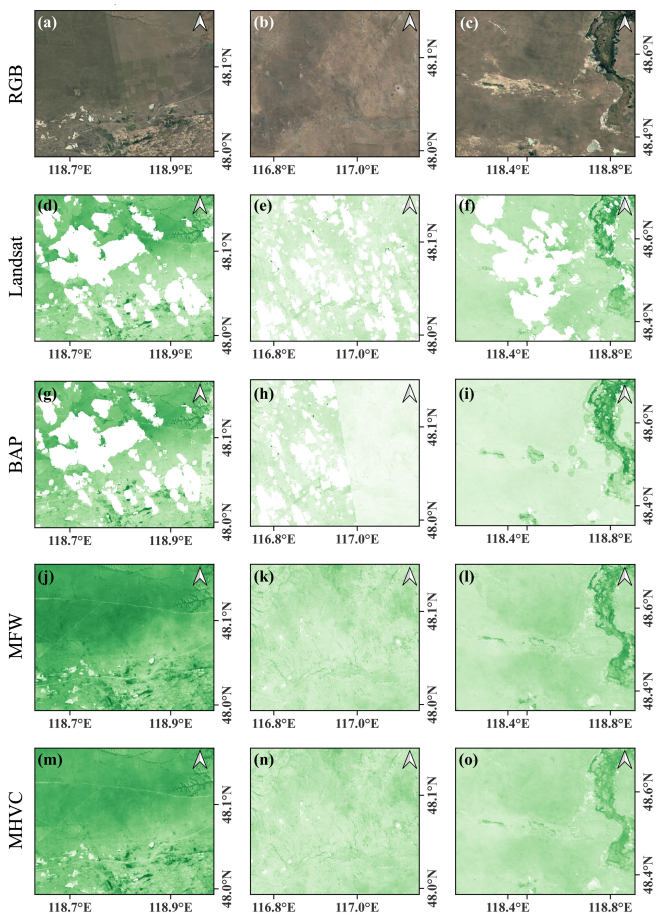


Fig. 6. NDVI compositing results over real cloud-contaminated Landsat scenes. (Columns from left to right) correspond to three acquisition dates: (a), (d), (g), (j), and (m) August 7, 2017, (b), (e), (h), (k), and (n) October 10, 2021, and (c), (f), (i), (l), and (o) June 4, 2024. (Rows from top to bottom) (a)–(c) RGB reference, (d)–(f) Landsat single-date image with cloud contamination, (g)–(i) BAP composite, (j)–(l) MFW composite, and (m)–(o) MHVC composite. White areas in (d)–(f) cloud- or cloud-shadow-masked pixels.

closely align with actual observations in terms of spatial detail and texture, demonstrating their robustness in complex surface environments.

To evaluate whether the visual advantages of MHVC extend beyond the summer growing peak, additional comparisons were performed for an early season date (August 7, 2017) and a late-season senescence date (October 10, 2021), shown in Appendix Fig. S1 and Fig. S2, respectively. Across both phenological stages, MHVC consistently reproduces fine-scale spatial structures, including dune ridge morphology and vegetation patch boundaries, with higher fidelity than BAP and MFW. These results indicate that the performance improvements observed for June 4, 2024, are not limited to a specific phenological stage but represent a robust structural characteristic of the framework throughout the full growing season.

2) Performance in Real Cloud-Contaminated Scenarios: Compositing results under real cloud-contaminated conditions across three phenological periods are illustrated in Fig. 6. The original Landsat scenes contain significant cloud and shadow contamination [Fig. 6(d)–(f)], appearing as irregular white patches that obscure surface features.

The BAP composite results [Fig. 6(g)–(i)] show noticeable residual noise and unreliable NDVI values due to incomplete cloud screening, resulting in fragmented spatial patterns that are particularly pronounced in the heavily contaminated August 7, 2017 scene [Fig. 6(g)]. The MFW method [Fig. 6(j)–(c)] improves spatial continuity in cloud-affected regions; however, it tends to underestimate NDVI values, weakening vegetation dynamic signals, as visible in the comparisons with the RGB reference [Fig. 6(a)–(c)].

In contrast, the MHVC method [Fig. 6(m)–(o)] reconstructs spatially complete NDVI distributions while maintaining consistent phenological characteristics across all three dates. By applying precise phenological control, MHVC effectively removes cloud contamination artifacts and preserves the fidelity of vegetation dynamics, outperforming both BAP and MFW in spatial coherence and spectral reliability.

3) Restoration Capability in Artificial Gap Simulations: To evaluate data restoration performance under controlled conditions, artificial data gaps of irregular shapes were simulated within high-quality reference areas, and gap-filling performance was quantitatively assessed against the true reference values (Fig. 7). Three mask geometries were applied across different phenological periods: a heart-shaped mask (Fig. 7(b), October 10, 2021), a star-shaped mask (Fig. 7(a), June 4, 2024), and a rectangular mask (Fig. 7(c), August 7, 2017).

As shown in Fig. 7(j)–(l), the MFW method inadvertently incorporates noncontemporaneous pixels during gap filling due to its broader SDOY window, leading to visible deviations in phenological characteristics and spectral tone within the reconstructed areas. This effect is particularly pronounced along the mask boundaries, where abrupt tonal transitions indicate inconsistencies between filled and surrounding pixels [Fig. 7(j)–(l)].

In contrast, MHVC provides the most reliable and spatially coherent reconstruction [Fig. 7(m)–(o)]. The filled areas integrate seamlessly with the surrounding environment in terms of both phenological traits and spatial textures, successfully reproducing the original surface features observed in the reference images [Fig. 7(d)–(f)]. These results demonstrate MHVC's superior gap-filling capability across seasons and mask geometries, highlighting its robustness in handling missing data under complex surface conditions.

4) Regional-Scale Spatial Continuity: At the regional scale (Fig. 8), the original Landsat image [Fig. 8(a)] is heavily affected by cloud contamination, leading to extensive data gaps. The BAP composite [Fig. 8(b)] exhibits pronounced brightness discontinuities and color inconsistencies at scene boundaries, compromising spatial coherence. The MFW composite [Fig. 8(c)] improves mosaicking consistency but still shows systematic large-scale distortions due to accumulated phenological mismatches. In contrast, MHVC [Fig. 8(d)] achieves the highest spatial coherence, with seamless scene transitions and well-preserved phenological gradients.

B. Quantitative Evaluation

1) Evaluation of Cross-Sensor Harmonization Performance: As illustrated in Fig. 9(a), the unadjusted Sentinel-2 NDVI exhibits a clear systematic bias relative to

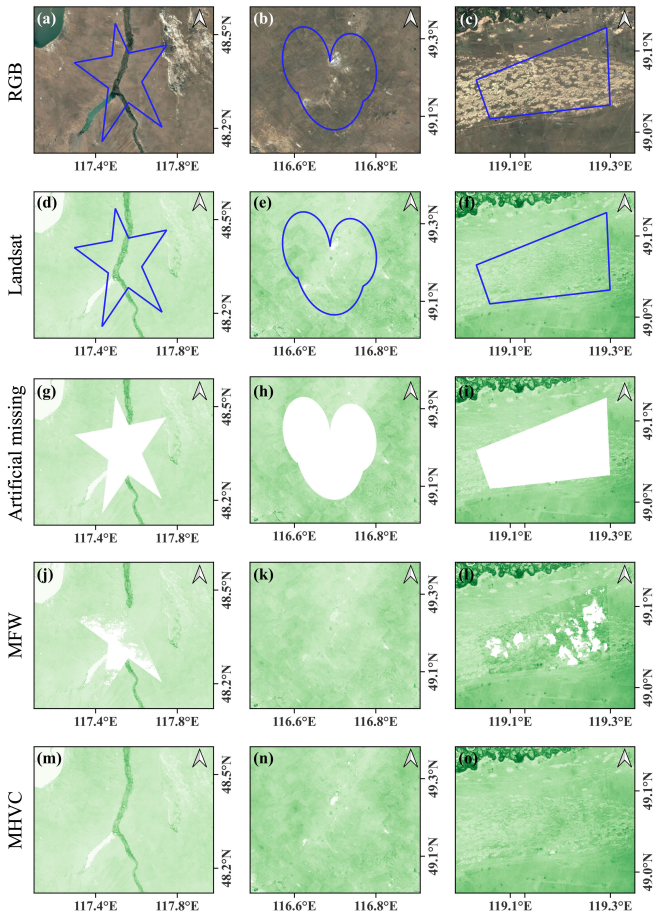


Fig. 7. Evaluation of gap-filling capability through artificial mask simulation. (Columns from left to right) study area, Area A, and Area B. (Rows from top to bottom) RGB true-color composite, original Landsat composite with the masked region outlined in blue, artificially masked image, MFW gap-filling result, and MHVC gap-filling result. Star-shaped mask (a), (d), (g), (j), and (m): applied to the study area of the Hulunbuir sandy land (Landsat reference acquired on June 4, 2024). Heart-shaped mask (b), (e), (h), (k), and (n): applied to Area A of the Hulunbuir sandy land (Landsat reference acquired on October 10, 2021). Rectangular mask (c), (f), (i), (l), and (o): applied to Area B of the Hulunbuir sandy land (Landsat reference acquired on August 7, 2017).

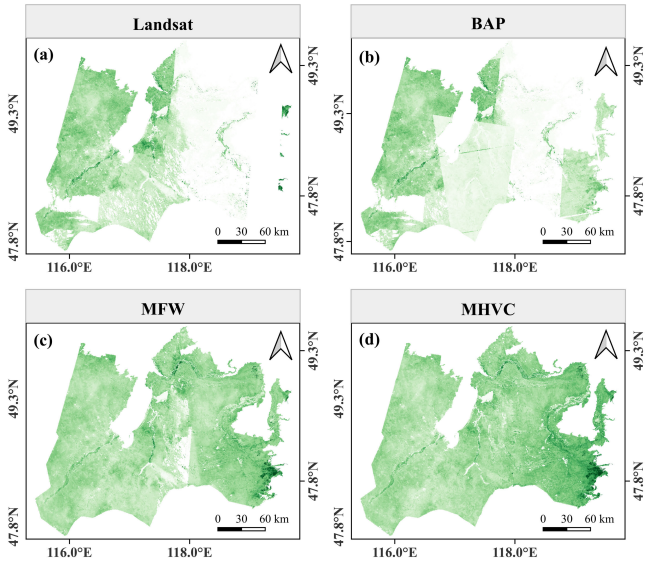


Fig. 8. Comparison of large-scale spatial continuity of NDVI composites across an extensive region in the Hulunbuir sandy land (Landsat reference acquired on October 10, 2021). The four panels show results from (a) Landsat single-date image, (b) BAP, (c) MFW, and (d) MHVC. White areas indicate cloud- or cloud-shadow-masked pixels.

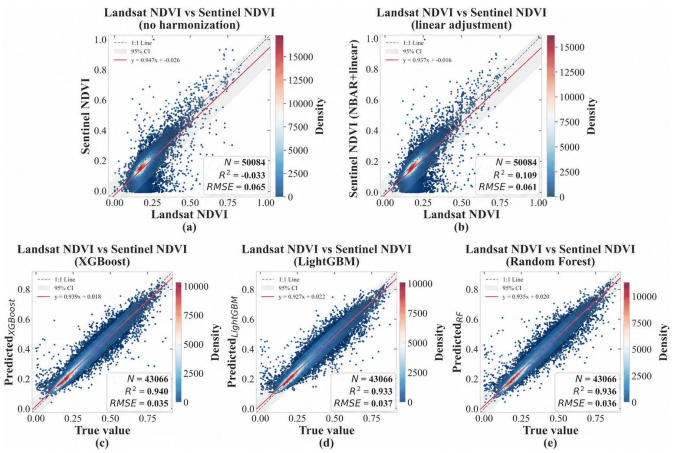


Fig. 9. Scatter plots of predicted versus reference Landsat NDVI. (a) Raw Sentinel-2 NDVI (no harmonization). (b) HLS 2.0 standard adjustment (RTLS BRDF normalization to NBAR + linear spectral correction). (c) XGBoost. (d) LightGBM. (e) Random forest.

Landsat-8/9 NDVI, with a negative offset of approximately 0.026 NDVI units and near-zero explained variance ($R^2 = -0.033$, $RMSE = 0.065$), indicating that raw Sentinel-2 reflectance is not directly interchangeable with Landsat-8/9 observations in heterogeneous sandy landscapes. The HLS 2.0 standard adjustment, which implements per-band BRDF normalization via the RossThick–LiSparse kernel-driven model using the official band-specific coefficients tabulated in [17], thereby converting Sentinel-2 surface reflectance to nadir BRDF-adjusted reflectance (NBAR), followed by the HLS 2.0 linear spectral band correction, provides only marginal improvement ($R^2 = 0.109$, $RMSE = 0.061$) [Fig. 9(b)], underscoring that even a physically grounded, standard two-step adjustment remains insufficient to capture the full complexity of cross-sensor discrepancy in this environment. The persistent scatter after HLS 2.0 correction suggests that residual discrepancies are driven by higher order interactions among solar zenith angle, observation geometry, surface anisotropy, and spatial context that cannot be adequately parameterized by a fixed kernel model with globally prescribed band coefficients. In contrast, the evaluated machine learning models demonstrate strong agreement with the 1:1 reference line. Among them, the XGBoost model achieves the highest precision ($R^2 = 0.940$, $RMSE = 0.035$) [Fig. 9(c)]. LightGBM and Random Forest also markedly outperform the HLS 2.0 adjustment ($R^2 = 0.933$, $RMSE = 0.037$ and $R^2 = 0.936$, $RMSE = 0.036$, respectively) [Fig. 9(d)–(e)]. These results confirm that machine learning approaches effectively capture nonlinear discrepancies driven by solar zenith angle, observation geometry, and spatial context [74].

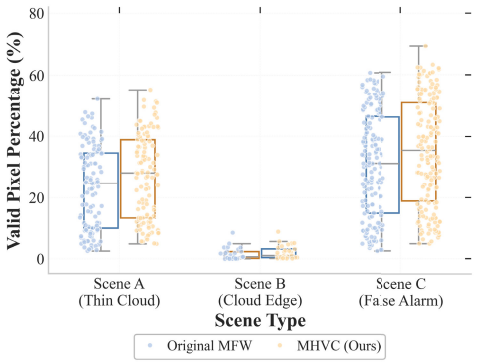


Fig. 10. Distribution comparison of valid pixel percentages between MFW and MHVC algorithms across typical scenarios.

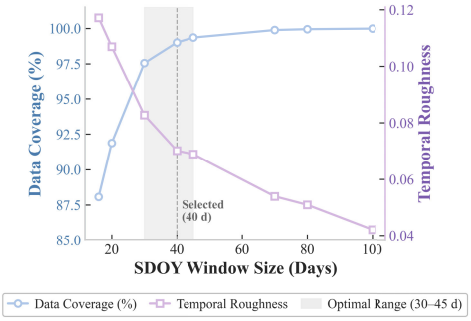


Fig. 11. Sensitivity of SDOY window size on NDVI compositing accuracy (coverage versus roughness).

2) Assessment of the MHVC Cloud Scoring Mechanism: The distance-probability joint scoring mechanism substantially increases the retention of valid pixels (Fig. 10), effectively alleviating the over-masking issues common in traditional geometric distance methods. In Scenario A (thin clouds/dust), MHVC achieves a markedly higher median proportion of valid pixels than the original MFW. In Scenario C (false alarms), MHVC exhibits a more compact distribution with higher valid pixel proportions, demonstrating its robustness in distinguishing high-reflectance sandy backgrounds from cloud shadows.

3) Optimization of the SDOY Compositing Window: The influence of the SDOY window size on compositing quality is shown in Fig. 11.

A systematic sensitivity analysis identifies 40 days as the optimal DOY compositing window for the Hulunbuir sandy land (Fig. 11). Short windows (< 30 days) produce suboptimal results. As shown by the purple curve, data coverage declines significantly (< 92%) while temporal roughness increases (> 0.10), suggesting that the composites are strongly influenced by anomalous observations and fail to bridge cloud-induced data gaps. In contrast, extending the window beyond 45 days leads to near-saturated coverage but substantially reduced temporal roughness, indicating potential over-smoothing, where the inclusion of too many noncontemporaneous observations masks genuine short-term phenological changes.

Consequently, a 40-day window is identified as the optimal balance. At this threshold, data coverage reaches 99.01% while temporal roughness (0.07) remains moderate, effectively

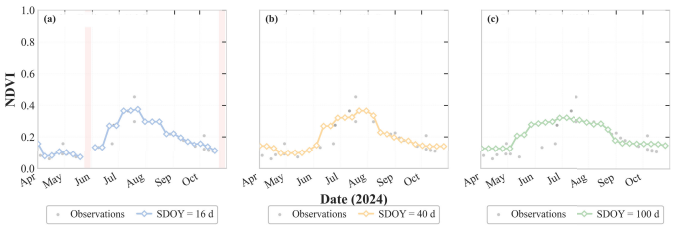


Fig. 12. Impact of SDOY window size on NDVI compositing quality and temporal smoothness. (a) Small window (16 days). (b) Optimal window (40 days). (c) Large window (100 days).

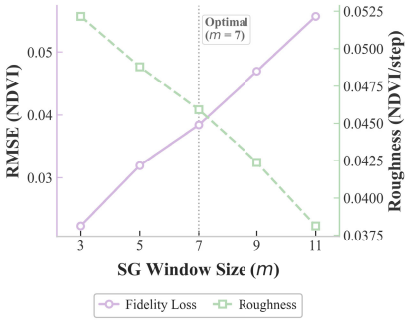


Fig. 13. Sensitivity analysis and parameter optimization for the S-G filter window size.

capturing phenological variability and yielding the highest compositing fidelity. In terms of effective temporal resolution, the MHVC framework generates one composite image per target date, with target dates spaced at eight-day intervals to align with the combined Landsat-8/9 nominal revisit cycle. The resulting time series thus provides an eight-day effective temporal resolution throughout the growing season (April–October), yielding approximately 26 composite epochs per year. This temporal density is sufficient to resolve the key phenological transitions of temperate sandy land vegetation, including spring green-up, peak biomass, and autumn senescence, as confirmed by the accurate extraction of SOS and EOS metrics demonstrated in Section V-B.

Fig. 12 visually validates this conclusion using typical vegetation pixel time-series trajectories. The short window (16 days, top panel) suffers from data gaps and jagged fluctuations due to cloud interference and satellite revisit constraints. The long window (100 days, bottom panel) eliminates voids but introduces severe step artifacts, flattening the true growth peaks by blending cross-seasonal data. The 40-day window (middle panel) delivers the best reconstruction, ensuring time-series continuity while preserving the fine dynamic features of the growing season, effectively avoiding both underfitting gaps and overfitting distortions.

4) Sensitivity Analysis of the Gf-S-G Filter: The selection of the S-G filter window size (m) is critical for NDVI time-series reconstruction (Fig. 13). The main objective is to balance fidelity loss, measured by RMSE, against temporal roughness. Sensitivity analysis across window sizes from 3 to 11 shows that small windows ($m < 5$) maintain low RMSE (<0.035) but exhibit high temporal roughness (>0.048), indicating insufficient suppression of residual noise.

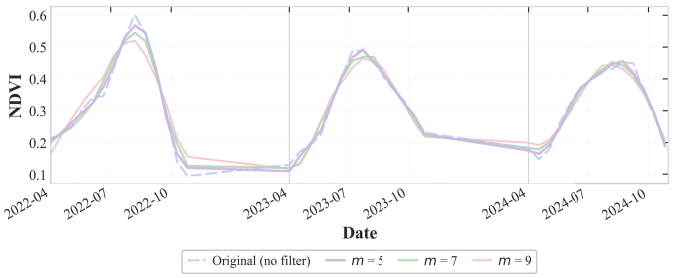


Fig. 14. Qualitative evaluation of filtering effects on a typical pixel NDVI time series (2022–2024).

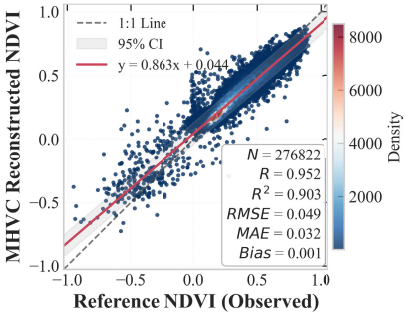


Fig. 15. Density scatter plot comparing reconstructed MHVC NDVI with observed reference values.

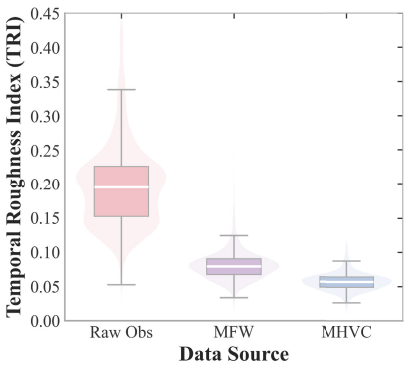


Fig. 16. Statistical distribution of TRI evaluating the temporal consistency of MHVC reconstructed products against raw observations and MFW.

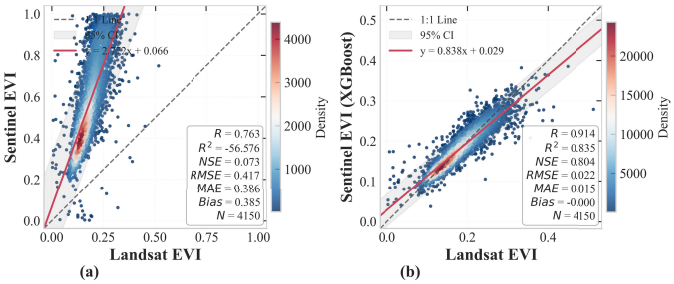


Fig. 17. Cross-sensor harmonization performance on EVI. (a) Raw Sentinel EVI. (b) XGBoost harmonized.

In contrast, larger windows significantly reduce roughness but increase RMSE, reflecting systematic deviations from the underlying signal. Based on the intersection of these metric curves, $m = 7$ is identified as the optimal window size, effectively suppressing noise while minimizing fidelity loss.

Time-series trajectories (Fig. 14) further corroborate these findings. A window size of 5 closely follows the original data but retains high-frequency, nonbiological jagged fluctuations. In contrast, a window size of nine produces an overly smooth curve, suppressing summer NDVI peaks and obscuring true vegetation dynamics. A window size of seven achieves the best performance, effectively reducing residual noise while preserving a smooth, continuous trajectory and accurately restoring the peak amplitude of the growing season.

5) Assessment of Temporal Consistency and Radiometric Integrity: To verify spectral fidelity, MHVC-reconstructed NDVI data (2017–2024) were evaluated pixel-by-pixel against high-quality Landsat reference observations (Fig. 15). The results reveal a strong linear correlation ($R^2 = 0.903$, $r = 0.952$) and low error metrics ($RMSE = 0.049$, $MAE = 0.032$). The near-zero bias ($= 0.001$) confirms the absence of systematic overestimation in either direction, demonstrating that the XGBoost harmonization and probability-weighted compositing introduce no directional radiometric drift into the reconstructed time series.

Furthermore, Fig. 16 displays the TRI distributions. The MHVC framework attains the lowest median TRI (≈ 0.06) compared to raw observations (≈ 0.20) and the MFW method (≈ 0.08), confirming its superior ability to generate spatiotemporally coherent continuous datasets with high stability. To further contextualize these quantitative results, it is instructive to examine the complementary roles played by the three MHVC components in driving the observed improvements. The reduction in median TRI from 0.20 (raw) to 0.08 (MFW) arises primarily from the weighted-averaging compositing strategy, which suppresses single-observation outliers. The additional reduction from 0.08 (MFW) to 0.06 (MHVC) reflects the combined effect of the probability-modulated cloud scoring, which increases the proportion of genuinely clear pixels entering the composite and thereby reduces the amplitude of residual contamination events, and the Gaussian-weighted S-G filter, which attenuates high-frequency noise while preserving phenological turning points. The near-zero bias (=0.001) in the spectral validation further indicates that neither the XGBoost harmonization nor the revised cloud-scoring introduces directional radiometric drift, a property that is critical for multiyear trend analysis and phenological parameter extraction in ecological monitoring applications.

V. Application Evaluation

A. Generalization to Other VIs

To evaluate the framework's generalization, the workflow was applied to another VI, the enhanced VI (EVI). As shown in Fig. 17, the raw Sentinel-2 EVI exhibits a clear systematic bias relative to Landsat-8 EVI, with a correlation of only 0.76. After applying the proposed harmonization adjustment, cross-sensor consistency improves substantially ($R^2 = 0.83$, $RMSE = 0.02$), accompanied by enhanced phase alignment and dynamic range fidelity. Because EVI incorporates the blue band and utilizes a more complex formula, its harmonization is inherently more challenging than NDVI, requiring slightly different feature weightings. Nevertheless,

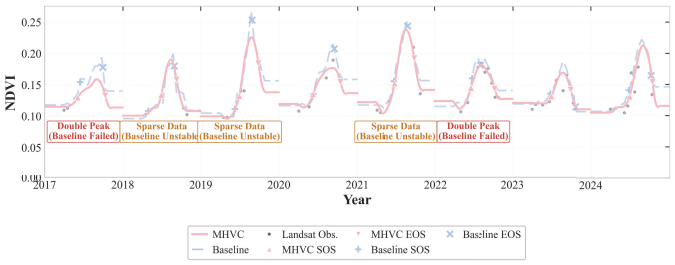


Fig. 18. Demonstration of interannual phenological dynamics monitoring using the MHVC framework (2017–2024) for a typical vegetation pixel.

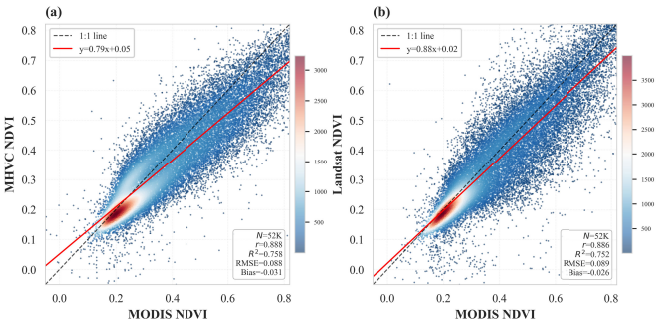


Fig. 19. Density scatter plots comparing spatially aggregated NDVI (250 m) against MODIS MOD13Q1 observations during the 2020–2024 growing seasons. (a) MHVC versus MODIS. (b) Landsat-8/9 versus MODIS.

the machine learning framework demonstrates strong adaptability and stability, attesting to its broad applicability across VI formulations and confirming that the multifeature design is not index-specific.

B. Validation of Phenological Metrics Extraction

Fig. 18 shows the long-term phenological trajectory (2017–2024) for a representative vegetation pixel (Point 865), demonstrating MHVC's robustness under complex conditions. The reconstructed trajectory (solid red line) closely follows Landsat reference observations (black scatter points), accurately reproducing the unimodal growth pattern of the vegetation.

The baseline MFW method (dashed blue line) exhibits notable limitations during data-sparse years (e.g., 2018, 2019, 2021), showing erratic, high-amplitude fluctuations. In addition, missing data during growth periods in 2017 and 2022 led MFW to produce biologically implausible "double-peak" patterns (double peak error), rendering it unreliable for phenological parameter extraction in these years.

MHVC effectively overcomes these limitations. Even in years susceptible to "double-peak" errors, it reconstructs a complete trajectory and accurately extracts key phenological metrics (SOS and EOS, indicated by upward and downward triangles). This robust performance under extreme data gaps demonstrates that the MHVC dataset provides a reliable foundation for fine-scale, interannual vegetation phenological monitoring at the regional level.

C. Independent Validation Against MODIS Time Series

To provide an independent assessment of the MHVC framework beyond the internal Landsat reference validation presented in Section IV-B5, reconstructed NDVI time series were evaluated against observations from the Moderate Resolution Imaging Spectroradiometer (MODIS) MOD13Q1 Version 6.1 product, a widely used benchmark for regional vegetation monitoring that is entirely independent of the Landsat–Sentinel-2 input data used in MHVC construction. MOD13Q1 provides 16-day composite NDVI at 250-m spatial resolution, with quality-controlled observations derived from Terra MODIS acquisitions. The validation period spans the 2020–2024 growing seasons (April–October), corresponding to the period of highest Sentinel-2 data density and most stable MHVC reconstruction performance. All MOD13Q1 pixels with SummaryQA flags of 0 (good quality) or 1 (marginal quality) were retained, while snow/ice and cloud-contaminated observations were excluded. MHVC outputs were spatially aggregated from 30 to 250 m through mean resampling with reprojection alignment to the MODIS grid, with a purity threshold requiring at least 80% valid pixel coverage within each aggregation window to minimize mixed-pixel contamination effects.

Temporal matching was performed by pairing each MOD13Q1 composite with the MHVC image whose target date fell within a ± 8 day window, yielding 66 matched image pairs across the five growing seasons. As shown in Fig. 19, MHVC NDVI exhibits strong agreement with MODIS observations overall ($r = 0.890$, $R^2 = 0.762$, RMSE = 0.087, Bias = -0.031). To contextualize these metrics, an equivalent comparison was conducted between spatially aggregated Landsat-8/9 NDVI, drawn from the same observation pool used as MHVC input, and the identical MODIS reference [Fig. 19(b)]. The Landsat-to-MODIS agreement ($r = 0.888$, $R^2 = 0.756$, RMSE = 0.088, Bias = -0.026) is virtually indistinguishable from that of MHVC, demonstrating that residual discrepancy with MODIS is attributable to the fundamental scale mismatch between 30-m Landsat/MHVC retrievals and 250-m MODIS composites rather than to errors introduced by the MHVC reconstruction process itself. This inference is consistent with established findings that spatial aggregation from fine to coarse resolution inherently compresses NDVI variance and reduces interproduct agreement, particularly in heterogeneous landscapes such as the sand–vegetation ecotone of the Hulunbuir sandy land. It is noteworthy that while the regression slope for MHVC ($y = 0.79x + 0.05$) is slightly lower than that of Landsat ($y = 0.88x + 0.02$), indicating modest compression of high NDVI values, this behavior is a physically expected consequence of the probability-weighted compositing strategy, which attenuates extreme observations in favor of temporally stable retrievals and does not represent a systematic bias in the reconstruction.

The systematic negative bias (Bias = -0.031) observed for MHVC is also physically interpretable. MOD13Q1 employs a constrained view angle maximum value compositing strategy, which tends to select higher NDVI observations within each compositing window and can thus yield slightly elevated values relative to time-targeted compositing methods such as MHVC. The small magnitude of this bias further confirms that the XGBoost harmonization model and

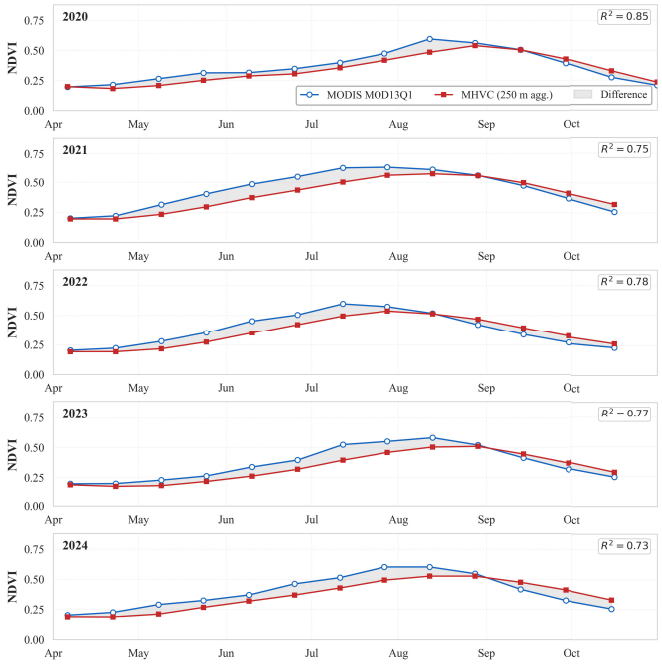


Fig. 20. Regional mean NDVI time-series comparison between MHVC (red) and MODIS MOD13Q1 (blue) for each growing season from 2020 to 2024. Shaded areas indicate the magnitude of interproduct differences. Annual R^2 values are annotated in the upper right of each panel.

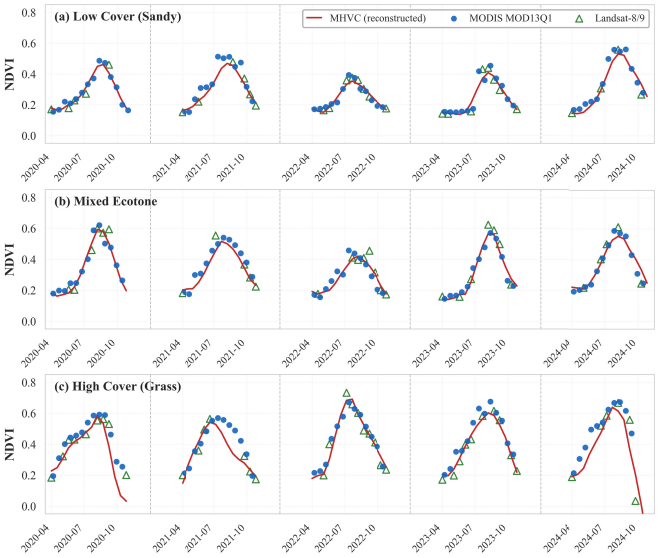


Fig. 21. Pixel-level NDVI time series at three representative sites spanning the vegetation cover gradient. (a) Low cover (sandy). (b) Mixed ecotone. (c) High cover (grass). The red line represents the MHVC SG-filtered reconstructed trajectory; green triangles indicate the original Landsat-8/9 acquisitions; and blue dots represent MODIS MOD13Q1 observations.

probability-weighted compositing pipeline do not introduce directional spectral distortions into the reconstructed time series.

At the regional scale, MHVC and MODIS exhibit closely coherent seasonal dynamics across all five years (Fig. 20). Both products capture the characteristic unimodal growing-season trajectory of the Hulunbuir sandy land, with peak NDVI typically occurring in late July to mid-August. Annual R^2 values between the regional mean time-series range from 0.73 (2024) to 0.85 (2020), indicating consistent phenological synchrony across years with varying hydrometeorological conditions. The slightly reduced agreement in 2024 ($R^2 = 0.73$) is likely attributable to an overall elevated NDVI amplitude in that year, in which a wider interproduct gap emerges near peak biomass, consistent with the tendency of MODIS maximum value compositing to amplify peak-season NDVI in years of strong vegetation signal. The gray shaded areas, representing the magnitude of interproduct differences, remain narrow throughout the green-up and senescence phases but widen modestly near peak biomass, reflecting the known tendency of MODIS to slightly overestimate peak-season NDVI relative to 30 m-based products in heterogeneous canopy environments.

At the pixel level, the trajectories of three representative sites spanning the vegetation cover gradient, a low-cover sandy site, a mixed ecotone, and a high-cover grassland pixel, are illustrated in Fig. 21. The MHVC reconstructed time series (red line) represents the SG-filtered continuous phenological trajectory; original Landsat-8/9 acquisitions are overlaid as discrete scatter points to illustrate the underlying observation density that supports the reconstruction. At all three sites, MODIS observations (blue dots) closely track the MHVC reconstructed time series throughout the growing season, with particularly strong agreement during the green-up acceleration phase (May–June) and the senescence stage (September–October). Minor discrepancies near the August peak at the high-cover site are consistent with the scale-induced variance compression discussed above and do not represent systematic errors in MHVC reconstruction. Taken together, these results confirm that MHVC time series are independently validated with respect to both spectral magnitude and phenological dynamics, providing additional confidence in the framework's suitability for regional vegetation monitoring applications.

D. Cross-Regional Generalizability Validation

To assess the spatial generalizability of the MHVC framework beyond the Hulunbuir sandy land, the complete reconstruction pipeline was applied to the remaining three major sandy lands in northern China: the Horqin sandy land, the Hunshandak sandy land, and the Mu Us sandy land. These regions collectively represent the full hydroclimatic gradient of temperate sandy ecosystems in China, ranging from the semi-humid agricultural–pastoral ecotone of Horqin (~300–450-mm mean annual precipitation) to the semi-arid shrub-dominated landscape of Hunshandak (~200–350 mm) and the arid dune systems of Mu Us (~150–400 mm), providing a rigorous testbed for framework transferability across divergent environmental conditions (Table I).

Given that comprehensive method comparisons against BAP and MFW have been conducted for the Hulunbuir sandy land (Section IV-A), the cross-regional evaluation focuses on the agreement between MHVC reconstructions and contemporaneous Landsat-8/9 reference observations to assess spatial generalizability across contrasting sandy land environments.

TABLE I

BASIC CHARACTERISTICS OF THE FOUR MAJOR SANDY LANDS USED FOR CROSS-REGIONAL GENERALIZABILITY VALIDATION

Sandy Land	Area (km^2)	MAP (mm)	Climate	Dominant Cover
Hulunbuir	47,600	350–400	Semi-humid	Grassland, fixed sand
Horqin	51,000	300–450	Semi-arid	Agro-pastoral mosaic
Hunshandak	29,000	200–350	Semi-arid	Shrub–grass, semi-fixed
Mu Us	39,000	150–400	Arid	Shrubland, semi-fixed

MAP: mean annual precipitation.

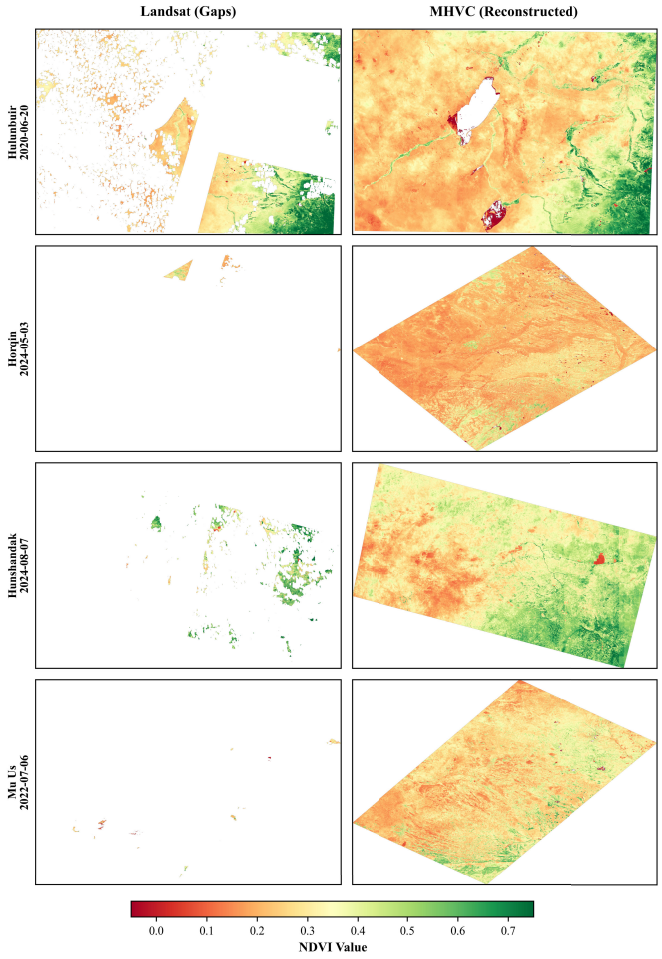


Fig. 22. Spatial data completeness comparison between raw Landsat-8/9 single-date acquisitions (left column) and MHVC reconstructed outputs (right column) for four major sandy lands under representative cloud-contaminated dates: Hulunbuir (June 20, 2020), Horqin (March 5, 2024), Hunshandak (August 7, 2024), and Mu Us (July 06, 2022). Gray areas indicate cloud- or cloud-shadow-masked pixels with no valid observations. Coverage percentages are annotated in the lower right corner of each panel. The color scale represents NDVI values ranging from −0.05 (red) to 0.75 (dark green).

1) Data Availability Recovery Across Regions: Fig. 22 illustrates the spatial distribution of NDVI under representative cloud-contaminated dates for all four sandy lands, contrasting raw Landsat-8/9 inputs with MHVC reconstructions. In all cases, the raw Landsat imagery suffers from severe data loss: valid pixel coverage ranges from as low as 0.2% (Mu Us, July 06, 2022) to 19.5% (Hulunbuir, June 20, 2020), leaving most of the study regions effectively unobservable from single-date Landsat acquisitions.

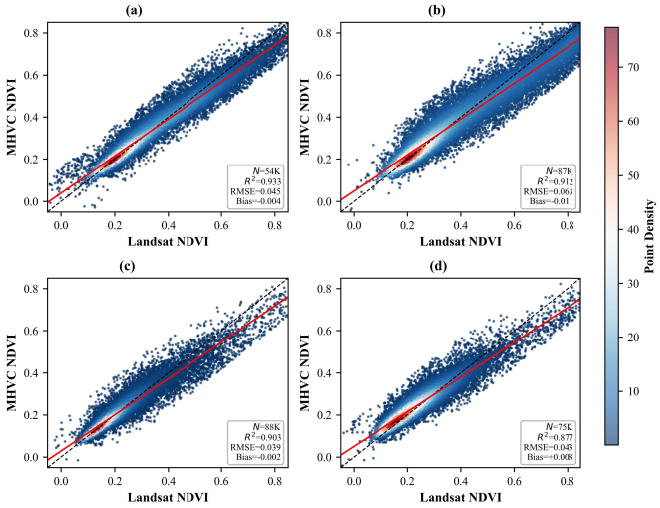


Fig. 23. Density scatter plots comparing MHVC-reconstructed NDVI with contemporaneous Landsat-8/9 reference observations for the four major sandy lands in northern China (a) Hulunbuir (N = 54K), (b) Horqin (N = 87K), (c) Hunshandak (N = 87K), and (d) Mu Us (N = 75K). Point colors represent kernel density estimates, with warmer tones indicating higher data concentration. The dashed line denotes the 1:1 reference, and the red solid line shows the linear regression. Statistical metrics are annotated in the lower right corner of each panel.

Following MHVC reconstruction, spatial coverage increases substantially across all four regions. For Hulunbuir, coverage rises from 19.5% to 94.3%, recovering nearly the full spatial extent of the study area. For Horqin, Hunshandak, and Mu Us, where near-total data loss (0.5%, 2.8%, and 0.2%, respectively) renders raw imagery essentially unusable for vegetation monitoring, MHVC reconstructs spatially coherent NDVI fields at 49.9%, 61.6%, and 47.9% coverage. The residual spatial gaps in the three new regions reflect the geometric footprint boundaries of the underlying Landsat scene tiles, which define the spatial extent of available observations within the compositing window, rather than residual cloud contamination. These results confirm that the probability-modulated compositing strategy effectively exploits all cloud-adjacent clear-sky observations across diverse surface and atmospheric conditions, delivering consistent data recovery regardless of regional climate regime.

2) Spectral Fidelity Across Regions: Fig. 23 presents density scatter plots of MHVC-reconstructed NDVI against contemporaneous Landsat-8/9 reference observations across all four sandy lands, and Table II summarizes the corresponding accuracy metrics. Across all regions, MHVC maintains strong spectral fidelity, with R^2 values ranging from 0.882 (Mu Us) to 0.935 (Hulunbuir) and Pearson correlation coefficients between 0.943 and 0.969. RMSE values remain consistently low, ranging from 0.039 (Hunshandak) to 0.063 (Horqin), and bias magnitudes do not exceed 0.011 across all regions, confirming the absence of systematic directional error in the reconstructed time series.

TABLE II
CROSS-REGIONAL ACCURACY OF MHVC RECONSTRUCTIONS AGAINST LANDSAT-8/9 REFERENCE OBSERVATIONS ACROSS THE FOUR MAJOR SANDY LANDS IN NORTHERN CHINA

Region	r	R^2	RMSE	MAE	Bias
Hulunbuir	0.969	0.935	0.044	0.032	−0.004
Horqin	0.959	0.912	0.063	0.047	−0.011
Hunshandak	0.953	0.905	0.039	0.025	−0.003
Mu Us	0.943	0.882	0.042	0.031	+0.007

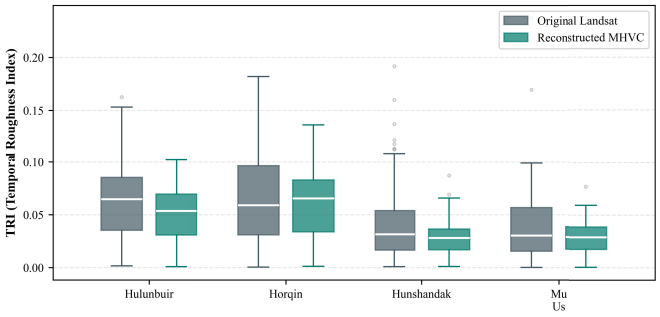


Fig. 24. Comparison of TRI distributions between raw Landsat-8/9 observations (light purple) and MHVC reconstructions (dark purple) across the four major sandy lands. Boxes indicate the interquartile range; horizontal lines denote the median; circles represent outliers beyond 1.5 × IQR. The slightly elevated MHVC TRI in Horqin reflects the inherently high phenological variability of its agricultural–pastoral mosaic landscape rather than reconstruction artifacts (see text for discussion).

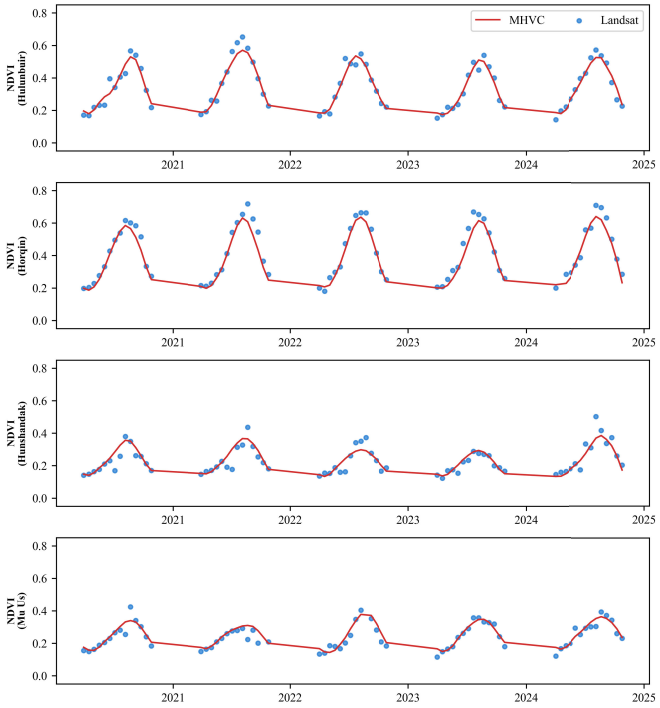


Fig. 25. Long-term NDVI time series (2020–2024) at representative vegetation pixels in the four major sandy lands (a) Hulunbuir (118.595°E, 48.847°N), (b) Horqin (120.380°E, 43.459°N), (c) Hunshandak (115.265°E, 42.516°N), and (d) Mu Us (109.510°E, 38.589°N). Red solid lines represent MHVC reconstructed trajectories; blue scatter points indicate contemporaneous Landsat-8/9 reference observations. Vertical dashed grid lines delineate calendar year boundaries.

The slight reduction in R^2 from Hulunbuir (0.935) to Mu Us (0.882) is physically interpretable. Mu Us is the driest of the four regions, with the lowest mean annual precipitation and the sparsest vegetation cover; under such conditions, the narrow NDVI dynamic range typical of arid sandy surfaces concentrates the majority of observations in the low-NDVI domain (0.10–0.30), where soil background reflectance variability and atmospheric residuals account for a proportionally larger fraction of the total signal variance [13], [59]. Conversely, Horqin's R^2 of 0.912 reflects the additional challenge posed by its complex agricultural–pastoral mosaic: rapid within-season NDVI transitions associated with crop cycles increase the effective phenological heterogeneity within any compositing window, slightly amplifying reconstruction uncertainty near the green-up and senescence transitions [36]. Importantly, the near-zero bias across all regions ($|\text{Bias}| \leq 0.011$) confirms that the XGBoost harmonization model does not introduce directional spectral offsets when applied beyond its original Hulunbuir training domain, a critical property for operational multiregion deployment.

3) Temporal Consistency Across Regions: Fig. 24 presents the TRI distributions for raw Landsat and MHVC reconstructions across all four sandy lands. For Hulunbuir and the two drier regions, Hunshandak and Mu Us, MHVC consistently reduces median TRI relative to raw Landsat observations: from ≈0.063 to ≈0.052 in Hulunbuir, from ≈ 0.035 to ≈0.028 in Hunshandak, and from ≈ 0.030 to ≈ 0.028 in Mu Us. These reductions confirm that the Gaussian-weighted S-G smoothing and probability-modulated compositing jointly suppress residual noise across contrasting climate regimes.

The pattern in Horqin warrants specific attention: median TRI for MHVC (≈0.065) slightly exceeds that of raw Landsat (≈0.058) in this region. This outcome is not a failure of the reconstruction framework but rather a direct consequence of the agricultural–pastoral mosaic characteristic of Horqin [4], [37]. In this region, raw Landsat observations are temporally sparse and preferentially available during clear-sky periods that often coincide with early or late phenological stages; as a result, the apparent smoothness of the raw time series partly reflects observation selection bias rather than genuine phenological continuity. MHVC, by recovering a denser and more temporally representative set of observations, including those capturing rapid crop green-up and senescence, produces a time series that more accurately reflects the true high-frequency phenological dynamics of this agro-pastoral ecotone, which are inherently more variable than those of the predominantly natural vegetation in Hulunbuir, Hunshandak, and Mu Us. This interpretation is consistent with the high spectral fidelity achieved in Horqin (R^2 = 0.912, $RMSE$ = 0.063), confirming that the slightly elevated TRI reflects real ecological dynamics rather than residual noise.

4) Phenological Dynamics at Representative Sites: Fig. 25 presents long-term NDVI time series (2020–2024) for representative vegetation pixels selected from each of the four sandy lands. At all four sites, the MHVC reconstructed trajectory

(solid red line) closely follows the Landsat-8/9 reference observations (blue scatter points), accurately capturing the unimodal seasonal pattern characteristic of temperate dryland vegetation.

At the Hulunbuir site (118.595°E, 48.847°N), MHVC faithfully reproduces both the peak-season NDVI amplitude (reaching ≈ 0.60–0.65 in July–August) and the interannual variability across all five growing seasons, with particularly close agreement between the reconstructed curve and the reference observations throughout the entire temporal extent. At the Horqin site (120.380°E, 43.459°N), the higher baseline NDVI (≈ 0.60–0.80 at peak) and more abrupt seasonal transitions reflect the agricultural–pastoral mosaic structure of this region. The MHVC trajectory captures the rapid green-up and senescence dynamics characteristic of cropland and pasture cycles, though minor step-like transitions are visible in some years, consistent with the compositionally heterogeneous nature of the landscape and the higher TRI values discussed above. At the Hunshandak site (115.265°E, 42.516°N) and Mu Us site (109.510°E, 38.589°N), MHVC accurately reconstructs the lower amplitude phenological trajectories (≈ 0.15–0.50) typical of semi-arid and arid sandy surfaces, with consistent tracking of the Landsat reference points across all five years. The near-zero NDVI values observed during the nongrowing season at these two sites are correctly reproduced, confirming that the S-G filter preserves the full dynamic range of the annual cycle without artificial elevation of winter baseline values.

Collectively, the cross-regional results demonstrate that the MHVC framework achieves consistent spectral fidelity (R^2 = 0.882–0.935), temporal stability (median TRI reduction confirmed in three of four regions), and phenological accuracy across all four major sandy lands in northern China, without requiring region-specific recalibration of any model parameter. This transferability is mechanistically grounded in the framework's design: by encoding geographic coordinates, multiscale spatial context, and seasonal angular variation as explicit model features, the XGBoost harmonization component learns spatially adaptive correction functions that generalize to new environments sharing similar structural characteristics, sparse canopy, bright soil background, and frequent atmospheric interference, regardless of specific geographic location or hydroclimatic regime.

VI. Discussion

A. Cross-Sensor Harmonization: Advantages of Machine Learning Over Linear Methods

The feature importance analysis of the XGBoost harmonization model (Fig. 26) provides mechanistic insight into the sources of intersensor NDVI discrepancy and elucidates why nonlinear approaches substantially outperform static linear calibration in this complex environment [67], [82].

As shown in Fig. 26(a), raw Sentinel-2 NDVI contributes the largest share of predictive power (85.4%), confirming that co-located spectral observation serves as the primary anchor for cross-sensor correction. However, the remaining 14.6% of predictive gain is distributed among geometric, spatial, temporal, and geographical features. This subset accounts for systematic intersensor discrepancies that spectral information alone cannot resolve [17], [79].

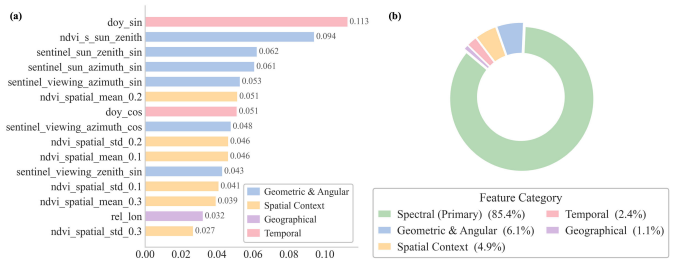


Fig. 26. Feature importance analysis of the XGBoost model (a) detailed ranking of auxiliary environmental and geometric variables and (b) major feature categories.

Among the secondary features, the geometric and angular group accounts for 6.1% of the total gain, with `doy_sin` and `ndvi_s_sun_zenith` emerging as the most influential predictors [Fig. 26(b)]. This highlights the significance of BRDF effects: although Landsat and Sentinel-2 have nominally similar overpass times (~10:00 and ~10:30 local time), differences in solar zenith angles produce systematic anisotropic reflectance offsets. These offsets are particularly pronounced during winter and over sparse canopy surfaces where directional reflectance contrast is maximized [78], [79]. The importance of viewing azimuth terms (`sentinel_viewing_azimuth_sin/cos`) further confirms that hotspot–darkspot geometry contributes additional variance in the sand–vegetation ecotone, consistent with findings by Claverie et al. [38], who emphasized that BRDF normalization is essential for robust intersensor harmonization.

Spatial context features account for 4.9% of the total gain, with `ndvi_spatial_mean_0.2` as the most influential predictor. This highlights the impact of mixed-pixel effects at landscape boundaries: subtle differences in sensor footprints and point spread functions lead to differential responses in pixels located at sand–vegetation transitions. The multiscale neighborhood design (0.1°–0.3°) enables the model to separate fine-scale patch variability from broader landscape gradients, a capability that is critical in the highly heterogeneous ecotone environment of the study area [56], [76], [82].

Temporal encoding features (`doy_sin`, `doy_cos`; 2.4% combined) enable season-aware correction by capturing the phenological modulation of intersensor discrepancy magnitude. During peak canopy development, angular differences induce larger NDVI offsets than during senescence or bare-soil periods [48], [78], [79]. Geographical features contribute a smaller share (1.1%), likely reflecting residual regional biases associated with aerosol gradients or sensor calibration drift across the satellite ground track [62].

Collectively, these results demonstrate that intersensor NDVI discrepancy is a multivariate interaction among illumination geometry, landscape heterogeneity, phenological stage, and geographic context. No single linear coefficient can adequately parameterize this interaction space, thereby explaining the substantial performance advantage of the machine learning framework over the HLS linear approach [79], [82]. This

finding is broadly consistent with recent advances in self-supervised and graph convolutional representation learning for remote sensing, including the self-supervised locality preserving low-pass graph convolutional embedding framework [25] and related hyperspectral feature extraction methods [26], [85], which have demonstrated that adaptive nonlinear embeddings substantially outperform fixed linear transformations in capturing spatially heterogeneous spectral structure in optically complex environments. Although those methods target hyperspectral image clustering rather than cross-sensor harmonization, the underlying principle, that high-dimensional spectral manifolds require nonlinear representations to be faithfully parameterized, directly supports the motivation for employing XGBoost over a globally invariant linear regression in the sandy land context.

B. Pixel Evaluation Mechanism Based on Distance–Probability Joint Constraints

Cloud and cloud shadow detection in sandy environments poses unique challenges that conventional distance-based scoring methods are ill-suited to address [36], [40], [46]. In traditional approaches such as MFW, pixel quality is penalized solely based on geometric proximity to the nearest detected cloud edge, a binary spatial logic that is fundamentally inadequate in arid and semi-arid regions for two reasons. First, the high surface reflectance of sandy soils can spectrally resemble clouds, frequently triggering false detections and causing valid clear-sky pixels to receive artificially low scores [39]. Second, frequent dust and aerosol events introduce continuous spectral ambiguity that deterministic cloud masks cannot reliably resolve, as spatial variability in dust optical depth cannot be captured by a fixed distance threshold [44].

The MHVC distance–probability joint scoring mechanism addresses these limitations by replacing fixed geometric penalties with probability-modulated weighting. When the confidence associated with a nearby flagged pixel is low, as is typical for thin clouds, dust plumes, or high-reflectance false alarms over bright sandy surfaces, the effective distance penalty applied to neighboring pixels is correspondingly reduced, allowing more valid observations to be retained during compositing [36], [40], [46]. This design is physically grounded in the probabilistic paradigm of uncertainty-aware cloud detection: a low-confidence cloud flag conveys fundamentally different information about contamination risk than a high-confidence detection, and treating them equivalently leads to unnecessary data loss that is most damaging precisely in the data-scarce environments where continuous time series are most needed.

The practical benefits are most pronounced in data-scarce arid environments. As shown in Fig. 27, across the top 15 high-gain scenes, MHVC recovers 13.3%–27.9% more valid pixels than MFW, with the largest improvement (+ 27.9%) observed in scenes from the most arid sampling units (T50UMU). This spatial pattern is consistent with the underlying mechanism: T50UMU tiles represent the driest areas of the study region, where bright sandy surfaces frequently induce false cloud detections and dust events are most persistent, conditions

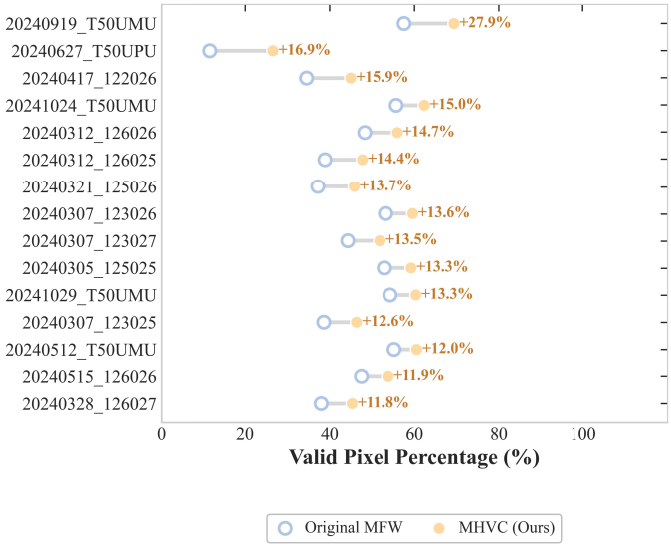


Fig. 27. Quantitative statistics of top 15 samples with the highest data recovery rate improvement via MHVC.

under which probability-modulated weighting offers a clear advantage over fixed-threshold approaches.

The cumulative distribution function (cdf) analysis in Fig. S3 further demonstrates that this advantage extends beyond individual high-gain scenes, representing a consistent improvement across the full distribution of valid pixel proportions. The MHVC curve exhibits a uniform rightward shift relative to MFW, with the shaded recovery gain area remaining positive across the entire range. This distributional robustness is critical as it indicates that probability weighting improves data availability systematically rather than being driven by a limited subset of anomalous scenes.

The downstream impact of this consistency is reflected in the reduced temporal roughness achieved by MHVC (median TRI $\approx$ 0.06 versus MFW $\approx$ 0.08; Fig. 16). This improvement arises not only from the application of the S-G filter but also from the increased proportion of valid, appropriately weighted observations entering the compositing process. When data recovery fails in individual years, the effects propagate beyond local gaps: limited valid observations force the reconstructed time series to rely more heavily on interpolation, amplifying residual noise and producing the spurious double-peak phenological artifacts observed in MFW outputs during data-sparse years (2017 and 2022; Fig. 18).

C. Bidirectional Gap-Filling Capability

Having established that MHVC achieves cross-sensor spectral consistency through machine learning-based harmonization and enhances data availability via probability-modulated cloud scoring, we now examine whether the framework supports true bidirectional gap-filling, specifically, whether Landsat observations can reliably substitute for Sentinel-2 during periods of missing data, complementing the Sentinel-to-Landsat substitution validated in Section IV-B5.

To evaluate this capability, bidirectional composite NDVI was compared pixel-by-pixel with contemporaneous

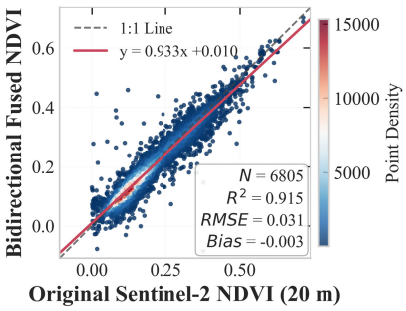


Fig. 28. Density scatter plot comparing bidirectional composite NDVI (Landsat → Sentinel-2) with original Sentinel-2 reference observations.

Sentinel-2 observations across 6805 sampling points during the 2024 growing season (April–October). As shown in Fig. 28, the results exhibit strong agreement with the Sentinel-2 observations (R^2 = 0.915, r = 0.957, $RMSE$ = 0.031, MAE = 0.019, $Bias$ = −0.003). These metrics are comparable to those obtained for the Sentinel-to-Landsat direction (R^2 = 0.903, $RMSE$ = 0.049; Section IV-B5), confirming symmetric performance in bidirectional gap filling.

From a mechanistic standpoint, the near-zero bias (−0.003) is particularly informative, indicating that the XGBoost harmonization model effectively removes systematic spectral offsets between Landsat and Sentinel-2 prior to gap-filling, thereby preventing directional radiometric drift in the reconstructed time series. The slight negative bias is physically consistent with the residual BRDF effects discussed in Section VI-A: Landsat's earlier overpass time results in marginally higher solar zenith angles at certain phenological stages, leading to a small but stable underestimation of Sentinel-equivalent NDVI that is only partially corrected by the model.

Temporal dynamics at representative sampling points (Appendix Fig. S4) further confirm that the bidirectional filling preserves phenological trajectory integrity across the vegetation cover gradient. At the low-cover sandy site (P3787, $RMSE$ = 0.019), the composite time-series closely follows the Sentinel-2 reference throughout the growing season, demonstrating strong model generalization in the low-NDVI domain characteristic of sand–vegetation ecotones. At the mixed ecotone site (P2862, $RMSE$ = 0.027), the composite series captures both the July green-up acceleration and the postpeak decline in September, with slight underestimation near the August peak attributable to the BRDF-induced systematic offset noted above.

At the high-cover vegetated site (P3236, $RMSE$ = 0.024), the composite series leads the Sentinel-2 reference slightly during the early green-up phase (May–June), likely reflecting the temporal weighting scheme, which assigns greater influence to Landsat observations capturing an earlier phenological state. This behavior is inherent to multisource compositing frameworks operating at coarser temporal sampling than the underlying phenological dynamics.

Taken together, these results demonstrate that MHVC enables robust bidirectional gap-filling with radiometric consistency and phenological fidelity across diverse surface conditions. The symmetry observed between the two filling directions is not coincidental but mechanistically grounded, arising from the same harmonization and probability-weighting components that underpin the framework. This confirms that cross-sensor consistency established during the harmonization stage propagates reliably into reconstructed time series, regardless of which sensor provides the gap-filling observations.

D. Limitations and Future Research Directions

Despite its computational efficiency, derived from the direct use of Level-2 surface reflectance products with embedded quality flags, which eliminates the need for repeated atmospheric correction and focuses processing on pixel quality assessment and temporal weighting, MHVC has several limitations that warrant consideration. Beyond the scale-dependency of interproduct agreement metrics discussed in Section IV-B5, three additional limitations merit explicit acknowledgment.

First, the machine learning-based harmonization model is trained to minimize overall prediction error across the training domain. As a result, its performance may degrade under spectral conditions that are underrepresented in the training data, reflecting a known limitation of empirical models in extrapolation scenarios. This issue is particularly relevant for interannual anomalies such as severe drought years or postfire recovery periods, during which surface spectral properties deviate substantially from typical growing-season conditions. Although the cross-regional validation presented in Section V-D demonstrates that the model generalizes effectively to Horqin, Hunshandak, and Mu Us without recalibration, performance in regions with substantially different soil spectral properties, such as dark volcanic substrates or carbonate-rich desert pavements, remains to be evaluated. Future work should explore domain-adaptive training strategies that explicitly account for interannual variability in surface conditions within the sandy land ecosystem.

Second, because Sentinel-2 observations are only available from 2017 onward, the framework cannot be directly applied to earlier periods, thereby limiting its utility for decadal-scale historical vegetation dynamics, a constraint common to all Landsat–Sentinel-2 compositing approaches. Extending the framework to incorporate Landsat 5 and 7 time series through cross-generational calibration represents a promising direction for overcoming this limitation, with the potential to enable continuous NDVI records dating back to the 1980s.

Finally, although the S-G filter effectively suppresses noise in the reconstructed time series, its performance is inherently constrained by the assumption of smooth, unimodal phenological trajectories, an assumption that may be violated in areas subject to abrupt disturbances such as grazing, fire, or flash flooding. In such settings, the filter may either under-correct high-frequency noise or inadvertently attenuate genuine short-duration phenological signals such as postrainfall green flushes. Incorporating synthetic aperture radar (SAR) data, which is insensitive to cloud cover and responsive to vegetation structural changes, could provide complementary gap-filling capacity under adverse weather conditions and facilitate reconstruction of disrupted phenological signals that cannot be resolved by optical-only frameworks.

In this study, we constructed a cross-sensor harmonization model by combining XGBoost with a carefully designed multisource feature system, achieving strong correction performance (R^2 = 0.94, $RMSE$ = 0.03). Through feature importance analysis, the framework successfully identifies the primary physical drivers of intersensor NDVI discrepancy, including vegetation cover fraction, phenological stage, and observation geometry, thereby maintaining a high degree of physical interpretability that is central to the scientific contribution of this work [86].

Meanwhile, we also recognize the considerable potential of deep learning in remote sensing data compositing. In recent years, deep learning methods have demonstrated notable advantages in multisensor fusion and time-series modeling [82], [87]. For instance, convolutional neural networks (CNNs) can automatically learn spatial context and higher order texture features directly from pixel neighborhoods, potentially offering superior characterization of complex boundary structures such as sand–vegetation transition zones [82], [87]; recurrent neural networks (RNNs/LSTM) or spatiotemporal Transformers can model long-range temporal dependencies in vegetation phenology in an end-to-end fashion, enabling more flexible handling of irregular observation sequences and abrupt phenological events as alternatives to the explicit DOY sine–cosine encoding currently employed [87], [88]; and self-supervised pretraining strategies can learn spectrally robust representations from large volumes of unlabelled imagery, helping to alleviate the scarcity of high-quality co-located Landsat–Sentinel-2 observation pairs and improving model generalization in observation-scarce regions [26], [89].

However, directly adopting a purely deep learning model in the specific context of this study may still encounter two principal challenges. First, the severe cloud contamination and sparse vegetation characteristic of sandy land environments result in a limited number of high-quality paired training samples, making it difficult to meet the large-scale data requirements typically associated with deep learning [82], [87]. Second, the inherent "opaque nature" nature of deep learning models reduces the interpretability of model decisions, increasing the difficulty of extracting clear physical mechanisms from feature contributions, whereas interpretability represents one of the core strengths of the present framework [86].

In future work, we plan to explore hybrid modeling approaches that incorporate lightweight CNN, Transformer, etc., branches as feature extraction components within the existing XGBoost framework. Such a design would retain the interpretability afforded by feature importance analysis while further enhancing the model's capacity to represent spatial context and complex temporal patterns, offering a principled path toward improved performance without sacrificing the operational tractability that makes the current framework suitable for large-area vegetation monitoring.

VII. Conclusion

This study presents the multifactor harmonized VI compositing (MHVC) framework to address the limitations of traditional MFW compositing methods in generating spatiotemporally continuous VI products from multisource remote sensing data in sandy regions. The framework was validated primarily over the Hulunbuir sandy land in northeastern China, a representative dryland ecosystem characterized by sparse observations, frequent dust interference, and rapid phenological transitions, and further evaluated across the Horqin, Hunshandak, and Mu Us sandy lands to assess spatial generalizability, demonstrating consistent improvements over existing approaches across four key dimensions.

First, MHVC employs a machine learning-based harmonization model to establish a nonlinear mapping between Landsat-8/9 and Sentinel-2 VIs prior to compositing, achieving substantially higher cross-sensor consistency (R^2 = 0.94, $RMSE$ = 0.03) than the conventional HLS linear adjustment method.

Second, a systematic sensitivity analysis identified an optimal ±20-day DOY compositing window, representing an effective balance between data availability and phenological fidelity.

Third, MHVC introduces a distance–probability joint scoring mechanism for cloud and shadow assessment, recovering up to 27.9% additional valid pixels in arid region samples compared to MFW.

Fourth, the GEE-based implementation operating directly on Level-2 surface reflectance products eliminates the need for repeated atmospheric correction across sensors and acquisition dates, enabling scalable large-area processing without additional preprocessing overhead. This design characteristic is validated through the cross-regional application to all four major sandy lands in northern China, covering a combined area of approximately 167 000 km^2, which was executed within the same computational framework without region-specific reconfiguration, confirming the operational scalability of the approach.

Quantitative validation against high-quality Landsat reference observations confirms that MHVC reconstructions achieve high spectral fidelity (R^2 = 0.903, $RMSE$ = 0.049, $Bias$ = 0.001) and superior temporal stability (median TRI ≈ 0.06). Independent validation against MODIS MOD13Q1 further demonstrates strong phenological consistency (r = 0.888), with discrepancies primarily attributable to scale differences rather than framework errors. Cross-regional application confirms robust transferability without region-specific recalibration.

Despite these advances, several limitations warrant acknowledgment. The framework relies on optical data and remains vulnerable to persistent heavy cloud cover or extreme dust events that simultaneously affect all observations within the compositing window. The XGBoost harmonization model, while demonstrated to generalize across four northern China sandy lands, has not been evaluated in regions with fundamentally different soil spectral properties, such as dark volcanic substrates or carbonate-rich desert pavements. The optimal 40-day compositing window may require adjustment in other environments: shorter windows (~20–30 days) are likely more appropriate for tropical drylands with rapid green-up dynamics, whereas longer windows may be needed in

boreal sandy regions with sparse observations [48], [51]; users are advised to conduct a sensitivity analysis analogous to Section IV-B3 before fixing this parameter in new regions. These limitations and the corresponding future research directions, including SAR integration for all-weather gap-filling, extension to historical Landsat archives, and domain-adaptive training, are discussed in detail in Section VI-D. Overall, MHVC provides a robust, phenologically faithful solution for vegetation monitoring in optically complex sandy land ecosystems and holds broad potential for global dryland applications.

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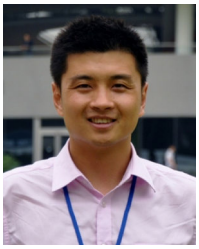
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